Your Group I, II, III and `4rho.tex`: every step checked; the terms $\propto \log \frac{k^+}{\Lambda}$ (JIMWLK) and $\propto \log \frac{\vee}{k^+}$ (BFKL), row by row; three- and four-ρ cancellation; the two-ρ sums and your two formulas

Notation as in your files: $\mathcal{K}(X) = X/X^2$, $K(a,b,z) = \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$, $U^{bc} = U^{cb}$,

$$\mathcal{N} = \frac{1}{(2\pi)^3}\frac{g^4}{16\pi^5}\frac{1}{k^+}.$$

Groups I and III are taken with their complex conjugate, Group II without, in every test (three ρ, two ρ, poles). Blue: new or corrected; red: removed. Only $\vee > k^+$ and $p^+ > \Lambda$ are used; $\Lambda \to 0$ wherever possible, $\vee$ exact, and every step is an exact identity. The remaining terms, the running coupling, DGLAP and the finite terms are in the companion file `Remaining_RC_DGLAP_finite.pdf`.

1 Answer

Your files. Every step of `Group_I.tex`, `Group_II.tex`, `Group_III.tex` and every equation of `4rho.tex` was checked numerically (Sec. 2). E1 (Group I Row V 2b UV) is now correct in your file. What is left to change:

M2 Group II Row IV, Eq. (1.16): phase of the second term (typo; the next equation is right).

4ρ `4rho.tex`, Eq. (1.12) (two ρ, first part): the factor $\left[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}\right]$ is missing (the earlier file had $\log\frac{\vee}{\Lambda}$); F1 and F2 are still needed in the two-ρ lists.

F3 Group I Row III d^3N_{3a}: the logarithmic line with the edge kernel is replaced by $\mathcal{D}$. Without it the three-ρ terms $\propto \log\frac{k^+}{\Lambda}$ do not cancel (0.59).

C5 Group III Row III part 2: the measure is $\int_{w,x',y',z,x}$ (M4).

M3, M5, M6 a prefactor sign in an unnumbered first line, a measure, typos (Sec. 2).

M2' Your JIMWLK formula: M at the wrong points and the sign of $dN^{(5)}$ (Sec. 11). Your BFKL formula is correct.

The split. Every row is split exactly, term by term,

$$\frac{d^3N_{\rm row}}{d^3k} = \log\frac{k^+}{\Lambda}\,\mathfrak{T}_{\rm row} + \log\frac{\vee}{k^+}\,\mathfrak{P}_{\rm row} + R_{\rm row}, \tag{1.1}$$

with $\log\frac{\vee}{\Lambda} = \log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}$ and the identities of Sec. 3. $R_{\rm row}$ (the remaining terms) is finite for $\vee \to \infty$ and $\Lambda \to 0$, which fixes $\mathfrak{T}$ and $\mathfrak{P}$ uniquely. Each of the 79 pointwise-evaluable terms equals its three parts to 1.5×10^{-14}.

Results.

- **Three** ρ**'s** $\propto \log\frac{k^+}{\Lambda}$ (22 terms): **zero** after symmetrization, 7.7×10^{-16} (Sec. 8).
- **Three** ρ**'s** $\propto \log\frac{\vee}{k^+}$ (35 terms): **zero**, 1.5×10^{-16}.
- **Four** ρ**'s** (`4rho.tex`, with $\left[\log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}\right]$): **zero** after symmetrization, 1.3×10^{-16} (Sec. 7). Their three- and two-ρ remainders enter both lists.
- **Two** ρ**'s** $\propto \log\frac{k^+}{\Lambda}$ = **JIMWLK⊗LO**, exactly: 13 kernel structures, 154 independent colour structures, 0 differences (Sec. 9). It is your JIMWLK formula $\sum_{i=1}^5 dN^{(i)}$ after the correction M2' (Sec. 11).
- **Two** ρ**'s** $\propto \log\frac{\vee}{k^+}$ = **BFKL⊗LO**, exactly: 9 kernel structures, 109 colour structures, 0 differences (Sec. 10). It is your BFKL formula, Eq. (LOBFKL), as you wrote it.

Therefore

$$\frac{d^3N_{\rm LL}}{d^3k} = \mathcal{N}\Big[\log\frac{k^+}{\Lambda}\,\text{JIMWLK}\otimes\text{LO} + \log\frac{\vee}{k^+}\,\text{BFKL}\otimes\text{LO}\Big] = -\mathcal{N}\log\frac{\Lambda}{k^+}\,\text{JIMWLK}\otimes\text{LO} + \mathcal{N}\log\frac{\vee}{k^+}\,\text{BFKL}\otimes\text{LO}. \tag{1.2}$$

In your formulas $\delta Y = \log\frac{k^+}{\Lambda}$ for JIMWLK (the target evolves from $y_T = \log\Lambda$ to the gluon) and $\delta Y = \log\frac{\vee}{k^+}$ for BFKL (the projectile evolves from the gluon to $y_P = \log\vee$). Written with $\log\frac{\Lambda}{k^+}$, the JIMWLK term has the coefficient $-$JIMWLK$\otimes$LO.

Where the BFKL terms are. In the explicit logarithms of the rows, and, in Group III Rows III, V, VI and Group II Row IV, inside line 1 (the p^+, resp. ξ, integral). There three exact identities bring them out: the subtraction of the plus prescription gives $-f(0)\log\frac{\vee}{k^+}$, which cancels the $\log\frac{\vee}{k^+}$ of line 2 exactly (I2); the Fourier transform of the y' integral at $y'=z$ gives the terms (B) (I3); the Fourier transform of $\log(y'-y)^2$ gives H_4 (I4). Nothing is expanded or dropped.

2 Your files, step by step

2.1 Group I, II, III

Every step that turns one of your equations into the next is checked numerically with your equations as written (read from your LaTeX): the p^+, q^+, ξ integrals, the w, w' integrals over $\delta^{(2)}$, the Wilson-line reductions, the splits (Group I Row V into 2a, 2b and 2b into UV and finite; Group II Row I part 2 into 2a–2f) and every plus prescription. Pointwise in all transverse coordinates, random points, random adjoint SU(3) Wilson lines ($U^{bc}=U^{cb}$), random charges, two configurations; remaining longitudinal integrals numerically. Equation numbers as your files compile. "plus": the plus-prescription steps hold up to $O(\Lambda)$; the two numbers are at $\Lambda=10^{-6}$ and 10^{-7}.

G	step	Eqs.	kind	rel. difference
I	Row II part 1: rewriting	(1.1) → (1.2)	integrals	0
I	Row III: p^+ integrals = N1+N2+N3a+N3b	(1.8) → (1.9), (1.10), (1.11), (1.12)	integrals	5.4×10^{-12}
I	Row IV: p^+ integral	(1.13) → (1.14)	integrals	5.2×10^{-16}
I	Row V part 1: w integral	(1.15) → (1.16)	fixed p^+	2.7×10^{-16}
I	Row V part 1: plus prescription	(1.16) → (1.17)	plus, $\Lambda\to0$	8.9×10^{-8} / 7.7×10^{-9}
I	Row V part 1: final form	(1.17) → (1.30)	integrals	0
I	Row V part 2: w integral	(1.18) → (1.19)	fixed p^+	4.3×10^{-16}
I	Row V part 2: rewriting	(1.19) → (1.20)	fixed p^+	2.7×10^{-15}
I	Row V part 2: identity $ffUU=N_cU$	(1.20) → (1.21)	fixed p^+	1.9×10^{-16}
I	Row V part 2: = 2a + 2b	(1.21) → (1.22), (1.23)	fixed p^+	5.1×10^{-16}
I	Row V 2a: plus prescription	(1.22) → (1.24)	plus, $\Lambda\to0$	1.6×10^{-7} / 1.3×10^{-8}
I	Row V 2a: final form	(1.24) → (1.31)	integrals	0
I	Row V 2b finite: simplifying ($q^+=k^+-p^+$)	(1.27) → (1.28)	integrals	1.9×10^{-16}
I	Row V 2b finite: plus prescription	(1.28) → (1.29)	plus, $\Lambda\to0$	8.7×10^{-8} / 7.4×10^{-9}
I	Row V 2b finite: final form	(1.29) → (1.33)	integrals	0
II	Row I part 1: p^+ integral	(1.1) → (1.2)	integrals	6.0×10^{-16}
II	Row I part 2: Wilson lines	(1.3) → (1.4)	fixed p^+	1.3×10^{-15}
II	Row I part 2: simplifying	(1.4) → (1.5)	fixed p^+	6.1×10^{-16}
II	Row I part 2: p^+ integral = 2a+...+2f	(1.5) → (1.6), (1.7), (1.8), (1.9), (1.10), (1.11)	integrals	6.1×10^{-16}
II	Row IV: w and w' integrals	(1.12) → (1.13)	fixed p^+	3.8×10^{-15}
II	Row IV: Wilson lines	(1.13) → (1.14)	fixed p^+	2.8×10^{-15}
II	Row IV: after the z integral → simplifying **(M2)**	(1.16) → (1.17)	fixed p^+	3.5×10^{-1}
II	Row IV: the same, phase of the 2nd term of Eq. (1.16) corrected	(1.16) → (1.17)	fixed p^+	1.9×10^{-16}
II	Row IV: $p^+\to\xi$	(1.17) → (1.18)	integrals	6.1×10^{-12}
II	Row IV: rewriting with $\mathcal{P}_{gg}$	(1.18) → (1.19)	integrals	1.1×10^{-15}
II	Row IV: plus prescription in ξ	(1.19) → (1.21)	plus, $\Lambda\to0$	3.8×10^{-7} / 3.0×10^{-8}
III	Row I: p^+ integral	(1.1) → (1.2)	integrals	1.5×10^{-16}
III	Row II: p^+ integral	(1.3) → (1.4)	integrals	1.7×10^{-16}
III	Row III part 1: w' integral	(1.5) → (1.6)	fixed p^+	1.1×10^{-15}
III	Row III part 1: plus prescription	(1.6) → (1.7)	plus, $\Lambda\to0$	1.1×10^{-6} / 8.8×10^{-8}
III	Row III part 2: w' integral	(1.8) → (1.9)	fixed p^+	1.2×10^{-15}
III	Row III part 2: plus prescription	(1.9) → (1.10)	plus, $\Lambda\to0$	1.3×10^{-7} / 1.1×10^{-8}
III	Row V: w' integral	(1.11) → (1.12)	fixed p^+	1.5×10^{-15}
III	Row V: plus prescription	(1.12) → (1.13)	plus, $\Lambda\to0$	1.1×10^{-6} / 8.8×10^{-8}
III	Row V: Wilson lines	(1.13) → (1.14)	integrals	2.9×10^{-14}
III	Row VI: w' integral	(1.15) → (1.16)	fixed p^+	2.1×10^{-15}
III	Row VI: plus prescription	(1.16) → (1.17)	plus, $\Lambda\to0$	1.1×10^{-6} / 8.8×10^{-8}

Group I Row V 2b, UV term (your E1 bracket): coefficient of the pole bracket vs. the exact p^+ integral, relative difference 7.5×10^{-16}.

Mistakes left in the Group files.

M2 Group II, Row IV, Eq. (1.16): in the second term the phase must be $e^{-ik\cdot(y'-y)\frac{p^++k^+}{k^+}}$, not $e^{-ik\cdot(y'-y)\frac{p^+}{(p^++k^+)}}$. A typo: Eq. (1.14) before and Eq. (1.17) after have the right phase. The step fails by 0.35 as written and agrees to 1.9×10^{-16} with the phase corrected.

M3 Group III, Row III, first (unnumbered) line: the prefactor is $-\frac{4}{(2\pi)^3}$, as in Rows V and VI ($-\frac{2}{(2\pi)^3}$ per term). Eqs. (1.5)–(1.10) below it are right.

M4 Group III, Row III, second part, Eqs. (1.8)–(1.10) (C5): the measure is $\int_{w,w',x',y',z,x}$ in Eq. (1.8) and $\int_{w,x',y',z,x}$ after the w' integral (Eqs. (1.9), (1.10)).

M5 Group I, Row II, Eq. (1.6): $\int_{w,w'} \to \int_{w,w',x'}$.

M6 Typos that do not change anything: Group II Eq. (1.13): $k\{+\} \to k^+$ in the phase. Group I Eq. (1.9): the opening `\Bigg\{` is never closed; Eqs. (1.19), (1.20), (1.22), (1.24): a closing `\Bigg\}` closes nothing. Group I Eqs. (1.16), (1.17), (1.30): $\int_{x,z,y,x',y'w'} \to \int_{x,z,y,x',y',w'}$; Eqs. (1.1), (1.2): a stray comma. Group II Row I: "We write it in two parts" twice.

The final forms of all rows (the ones used below) are the same as in the files of the earlier checks, apart from your corrected E1 bracket: their log $\frac{k^+}{\Lambda}$ parts, log $\frac{\vee}{k^+}$ parts and remainders agree row by row and pointwise (0, `same_lists.py`). Your earlier `BFKL.tex` + `Other_terms.tex` are these rows minus their JIMWLK parts ($\le 2.5 \times 10^{-9}$, i.e. $O(\Lambda)$ at $\Lambda = 10^{-9}$; `rows_vs_files.py`).

2.2 `4rho.tex`

Each equation against the earlier, checked `4rho.tex`, pointwise (random charges, points and Wilson lines; log $\frac{\vee}{\Lambda}$ there, $\left[\log \frac{k^+}{\Lambda} + \log \frac{\vee}{k^+}\right]$ here). The two equations written as "prefactor $\times\rho\rho\rho\rho[A] + [B]$" are compared as text.

Eq.		new vs. earlier file
(1.1)	four ρ's, Eq. (4rho)	5.8×10^{-16}
(1.2)	its symmetrized sum $= 0$	both 0
(1.3)	four Wilson lines, rewritten	identical text (without the log bracket); logs 3/3
(1.4)	four Wilson lines, $= 0$	identical text (without the log bracket); logs 1/1
(1.5)	three ρ: first form	4.0×10^{-16}
(1.6)	three ρ	4.1×10^{-16}
(1.7)	three ρ	4.1×10^{-16}
(1.8)	three ρ	2.1×10^{-16}
(1.9)	three ρ	3.1×10^{-16}
(1.10)	three ρ	3.1×10^{-16}
(1.11)	three ρ, $= 0$	both 0
(1.12)	two ρ, first part	**ratio 0.0951** $= 1/\log \frac{\vee}{\Lambda}$: the log bracket is missing
(1.13)	two ρ, second part (1)	identical text (without the log bracket); logs 1/1
(1.14)	two ρ, second part (2)	6.1×10^{-16}

- **The four-ρ terms cancel** after symmetrization over the four charges: residual / size 1.3×10^{-16} (three configurations; `three_rho/four_check.py`, whose transcription equals your Eq. (4rho)).
- **Two ρ, first part, Eq. (1.12):** the prefactor is $\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}$; it must be $\frac{g^4}{8\pi^3}\frac{\left[\log \frac{k^+}{\Lambda} + \log \frac{\vee}{k^+}\right]}{2\pi^2}$, as in the second part. As written it is a log-free (finite) term, and the evolution would miss it.
- **The two-ρ lists** vs. the checked lists (Final_evolution_check.pdf, Eq. (fourlists)), charges symmetrized, with c.c., three configurations: with the log restored and F1, F2: 5.6×10^{-15}; with the log restored and F2 only: 0.07–1.37; without the log (F2 only): 0.46–0.99. F2 is always needed: as written, the index b' occurs four times in one term.

F1 first part: drop $-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)\,\rho\rho\,UU$ (the term starting with -8);

F2 second part, $dN^{(1)}$, term 2: $f^{abc}f^{deg}\rho^{b'}(x')\rho^c(y)U^{ag}(y)U^{db'}(w')U^{eb}(z)$.

3 How the logarithms are split: exact identities

(I1) Explicit logarithms. A term carries at most one logarithmic factor. Your brackets:

$$\left[\log\left(\frac{k^+}{\Lambda}\right) + \log\left(\frac{\vee}{k^+} - 1\right)\right] = \log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+} + \log\left(1 - \frac{k^+}{\vee}\right),$$
$$\left[\log\left[\frac{\vee}{k^+}\right] + \log\left[\frac{k^+}{\Lambda}\right]\right] = \log\frac{k^+}{\Lambda} + \log\frac{\vee}{k^+}.$$

All other logarithms ($A = (x-z)^2$ etc.; the remainder exactly and, where it depends on Λ, its limit):

$$\log\left(\frac{\vee}{k^+} - 1\right) = \log\frac{\vee}{k^+} + \log\left(1 - \frac{k^+}{\vee}\right)$$
$$\log\left(\frac{k^+ - \Lambda}{\Lambda}\right) = \log\frac{k^+}{\Lambda} + \log\left(1 - \frac{\Lambda}{k^+}\right)$$
$$\xrightarrow{\Lambda\to 0} \log\frac{k^+}{\Lambda}$$
$$\log\left[\frac{k^+ - \Lambda}{\Lambda}\right] = \log\frac{k^+}{\Lambda} + \log\left(1 - \frac{\Lambda}{k^+}\right)$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{k^+}{\Lambda}$$

$$\log\left[\frac{\Lambda}{k^+-\Lambda}\right]=-\log\frac{k^+}{\Lambda}-\log\left(1-\frac{\Lambda}{k^+}\right)$$

$$\xrightarrow{\Lambda\to 0}\ -\log\frac{k^+}{\Lambda}$$

$$\log\left[\frac{\vee}{\Lambda+k^+}\right]=\log\frac{\vee}{k^+}-\log\left(1+\frac{\Lambda}{k^+}\right)$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}$$

$$\log\left[\frac{\vee}{k^++\Lambda}\right]=\log\frac{\vee}{k^+}-\log\left(1+\frac{\Lambda}{k^+}\right)$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}$$

$$\log\left[\frac{\left[\vee\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[(\Lambda+k^+)\,(x-z)^2-k^+\,(x-w)^2\right]}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{\left(1+\frac{\Lambda}{k^+}\right)(x-z)^2-(x-w)^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}+\log\left[\frac{(x-z)^2-\frac{k^+}{\vee}\,(x-w)^2}{(x-z)^2-(x-w)^2}\right]$$

$$\log\left[\frac{\left[(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2\right]}{\left[\Lambda\,(x-z)^2-k^+\,(x-w)^2\right]}\right]=\log\left[\frac{(k^+-\Lambda)\,(x-z)^2-k^+\,(x-w)^2}{\Lambda\,(x-z)^2-k^+\,(x-w)^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$

$$\log\left[\frac{\left[k^+\,(x-w)^2-\vee\,(x-z)^2\right]}{\left[k^+\,(x-w)^2-\Lambda\,(x-z)^2\right]}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2}{(x-w)^2-\frac{\Lambda}{k^+}\,(x-z)^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}+\log\left[\frac{\frac{k^+}{\vee}\,(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$

$$\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,\vee}{(x-w)^2\,k^+-(x-z)^2\,(k^++\Lambda)}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2\left(1+\frac{\Lambda}{k^+}\right)}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}+\log\left[\frac{(x-w)^2\,\frac{k^+}{\vee}-(x-z)^2}{(x-w)^2-(x-z)^2}\right]$$

$$\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]=\log\left[\frac{(x-w)^2\,k^+-(x-z)^2\,(k^+-\Lambda)}{(x-w)^2\,k^+-(x-z)^2\,\Lambda}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\left[\frac{(x-w)^2-(x-z)^2}{(x-w)^2}\right]$$

$$\log\left[\frac{\left[k^+\,(x-w)-(k^+-\Lambda)\,(x-z)\right]^2}{\left[k^+\,(x-w)-\Lambda\,(x-z)\right]^2}\right]=\log\left[\frac{\left[k^+\,(x-w)-(k^+-\Lambda)\,(x-z)\right]^2}{\left[k^+\,(x-w)-\Lambda\,(x-z)\right]^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\left[\frac{(z-w)^2}{(x-w)^2}\right]$$

$$\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{\Lambda\,(x-z)^2+k^+\,(x-w)^2}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]$$

$$\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w)^2}{k^+\,(x-w)^2+\Lambda\,(x-z)^2}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w)^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w)^2}{(x-w)^2}\right]$$

$$\log\left[\frac{(\vee-k^+)\,(x-z)^2+k^+\,(x-w')^2}{\Lambda\,(x-z)^2+k^+\,(x-w')^2}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{\frac{\Lambda}{k^+}\,(x-z)^2+(x-w')^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}\,(x-w')^2}{(x-w')^2}\right]$$

$$\log\left[\frac{(\vee-k^+)\,(x'-z)^2+k^+\,(x'-w')^2}{\Lambda\,(x'-z)^2+k^+\,(x'-w')^2}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}\,(x'-w')^2}{\frac{\Lambda}{k^+}\,(x'-z)^2+(x'-w')^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{(x'-w')^2}\right]$$

$$\log\left[\frac{k^+(x'-w')^2+(\vee-k^+)(x'-z)^2}{k^+(x'-w')^2+\Lambda(x'-z)^2}\right]=\log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{\frac{\Lambda}{k^+}(x'-z)^2+(x'-w')^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\frac{\vee}{k^+}+\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x'-z)^2+\frac{k^+}{\vee}(x'-w')^2}{(x'-w')^2}\right]$$

$$\log\left[\frac{(\vee-k^+)(x-z)^2+k^+(x-w)^2}{k^+(x'-w')^2+(\vee-k^+)(x'-z)^2}\right]=\log\left[\frac{\left(1-\frac{k^+}{\vee}\right)(x-z)^2+\frac{k^+}{\vee}(x-w)^2}{\frac{k^+}{\vee}(x'-w')^2+\left(1-\frac{k^+}{\vee}\right)(x'-z)^2}\right]$$

$$\log\left[\frac{\Lambda(x-z)^2+k^+(x-w)^2}{k^+(x'-w')^2+\Lambda(x'-z)^2}\right]=\log\left[\frac{\Lambda(x-z)^2+k^+(x-w)^2}{k^+(x'-w')^2+\Lambda(x'-z)^2}\right]$$

$$\xrightarrow{\Lambda\to 0}\ \log\left[\frac{(x-w)^2}{(x'-w')^2}\right] \tag{3.1}$$

Logarithms of ϵ, μ, $\log(1-\frac{k^+}{\vee})$ and $\log(1-\frac{\Lambda}{k^+})$ are finite and are not split. Three special cases:

- Group I Row II part 1: $\left[\frac{11}{3}-2\log\frac{k^+}{\Lambda}+\log(1-\frac{k^+}{\vee})\right]$ times the pole bracket; only $-2\log\frac{k^+}{\Lambda}$ is large.
- Group I Row V 2b UV: $\left[2\log\frac{k^+}{\Lambda}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}\right]$.
- Group I Row V 2b finite, line 6: $\left[-\frac{11}{6}+2\log\frac{k^+-\Lambda}{\Lambda}\right]$.

(I2) The subtraction of the plus prescription. With your $\left[\frac{1}{p^+}\right]_+ f=\frac{f(p^+)-f(0)}{p^+}$, define $\left[\frac{k^+}{p^+(p^++k^+)}\right]_+ f\equiv\frac{k^+[f(p^+)-f(0)]}{p^+(p^++k^+)}$. Since $\frac{1}{p^+}=\frac{1}{p^++k^+}+\frac{k^+}{p^+(p^++k^+)}$ and $\int_0^{\vee-k^+}\frac{dp^+}{p^++k^+}=\log\frac{\vee}{k^+}$,

$$\int_0^{\vee-k^+}dp^+\left[\frac{1}{p^+}\right]_+ f(p^+)=\int_0^{\vee-k^+}dp^+\left(\left[\frac{k^+}{p^+(p^++k^+)}\right]_+ +\frac{1}{p^++k^+}\right)f(p^+)-f(0)\log\frac{\vee}{k^+}. \tag{3.2}$$

For ξ, with your $\frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}g=\frac{d\xi}{\xi(1-\xi)}[g(\xi)-g(1)]$, define $\frac{d\xi}{\xi^2}\frac{\xi^2}{(1-\xi)_+}g\equiv\frac{d\xi}{1-\xi}[g(\xi)-g(1)]$. Then $\frac{1}{\xi(1-\xi)}=\frac{1}{\xi}+\frac{1}{1-\xi}$ gives

$$\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\frac{\xi}{(1-\xi)_+}g=\int_{\frac{k^+}{\vee}}^1\frac{d\xi}{\xi^2}\Big(\xi+\frac{\xi^2}{(1-\xi)_+}\Big)g-g(1)\log\frac{\vee}{k^+}. \tag{3.3}$$

Your line 2 is $f(0)\left[\log\frac{k^+}{\Lambda}+\log(\frac{\vee}{k^+}-1)\right]$ (resp. $g(1)[\dots]$): it came from this subtraction. So its $\log\frac{\vee}{k^+}$, (i), and $-f(0)\log\frac{\vee}{k^+}$, (ii), cancel exactly, poles included; $\log\frac{k^+}{\Lambda}$ (JIMWLK) and $\log(1-\frac{k^+}{\vee})$ (remaining) are left.

(I3) The y' integral at $y'=z$. In line 1 of Group III Rows III(1), V, VI take the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term; in Row III(2) the $-\frac{p^+}{k^{+2}}$ term. Its coefficient tends to a constant at large p^+. Write

$$G^{k'}(y')=\frac{(x'-y')^{k'}}{(x'-y')^2}\times\left[\text{colour bracket with } U(y')\right]=G^{k'}(z)+\left[G^{k'}(y')-G^{k'}(z)\right].$$

With $\int_u e^{-iq\cdot u}\frac{u^m}{u^2}=-2\pi i\frac{q^m}{q^2}$ and $q=k\frac{p^++k^+}{k^+}$, exactly,

$$\int_{y'}e^{-ik\cdot(y'-w)}e^{-ik\cdot\frac{p^+}{k^+}(y'-z)}\frac{(y'-z)^m}{(y'-z)^2}\,G^{k'}(z)=\frac{k^+}{p^++k^+}\int_{w'}e^{-ik\cdot(w'-w)}\frac{(w'-z)^m}{(w'-z)^2}\,G^{k'}(z). \tag{3.4}$$

For Rows III(1), V, VI: $\int_0^{\vee-k^+}\frac{dp^+}{2\pi}\left(-\frac{1}{k^+}\right)\frac{k^+}{p^++k^+}=-\frac{1}{2\pi}\log\frac{\vee}{k^+}$, exactly. This is (B), (iii), with nothing left over. The $G(y')-G(z)$ part stays in the remaining terms.

(I5) Row III(2). With $A=(x-z)^2$, $B=(x-w)^2$,

$$-\frac{p^+}{k^{+2}}\frac{k^+}{p^++k^+}\frac{1}{p^+A+k^+B}=-\frac{1}{k^+A(p^++k^+)}+\frac{B}{A(p^++k^+)(p^+A+k^+B)}. \tag{3.5}$$

The first term gives (B). The second falls like $\frac{1}{p^{+2}}$ and stays in the remaining terms.

(I4) Group II Row IV. In the $\frac{1-\xi}{\xi}$ term of line 1 write $H(y,y') = H(y,y) + [H(y,y') - H(y,y)]$, $H(y,y') = \frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}[U^{e'c}(y')-U^{e'c}(x')][U^{ec}(y)-U^{ce}(x)]\rho^{e'}(x')\rho^{e}(x)$. The constant part of the pole bracket gives $\delta^{(2)}(k/\xi) = 0$ for $k \neq 0$, and $\int_{y'} e^{-ik\cdot(y'-y)\frac{1}{\xi}} \log\frac{(y'-y)^2\mu^2}{4} = -\frac{4\pi\xi^2}{k^2}$. Also $\int_{\frac{k^+}{\vee}}^1 d\xi\,\frac{1-\xi}{\xi} = \log\frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}$ and $\int_{w,w'} e^{-ik\cdot(w'-w)}\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z) = \frac{4\pi^2}{k^2}$. So the $H(y,y)$ part is, exactly,

$$\frac{g^2}{4\pi^2}\frac{N_c}{(2\pi)^3}\frac{g^2}{2\pi^2}\frac{1}{k^+}\frac{d_\perp}{2}\frac{4\pi}{k^2}\left[\log\frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}\right]\int_{x,x',z} H(z,z) = \left[\log\frac{\vee}{k^+} - 1 + \frac{k^+}{\vee}\right] H_4\Big|_{\log\frac{\vee}{k^+}\to 1}, \tag{3.6}$$

at $d_\perp = 2$ (no pole multiplies it). Its $\log\frac{\vee}{k^+}$ term is $\log\frac{\vee}{k^+} H_4$, (iii) of Group II Row IV.

The split is complete. After these steps every remaining p^+ (ξ) integrand falls at least like $\frac{1}{p^{+2}}$ after the transverse integrals ($G(y') - G(z)$ and $H(y,y') - H(y,y)$ vanish at the coincident point; the $\frac{1}{p^+ + k^+}$ and $\left[\frac{k^+}{p^+(p^+ + k^+)}\right]_+$ terms carry the $\frac{k^+}{p^+ + k^+}$ of (I3)). So the remaining terms stay finite for $\vee \to \infty$, and the coefficient of $\log\frac{\vee}{k^+}$ is unique; $\vee$ stays exact in the remaining terms.

Numerical checks (`bfkl_rows/exact_check.py`; two configurations, $\vee = 20k^+$, the p^+ and ξ integrals done numerically, pointwise in the transverse points; line 1 of these rows is the same in your new files):

- line 1 = (rewritten line 1) + ($G(y') - G(z)$ term) + $(-f(0)\log\frac{\vee}{k^+})$ + ($G(z)$ term), Group III Rows III(1), III(2), V, VI; line 1 = (rewritten line 1) + $(-g(1)\log\frac{\vee}{k^+})$, Group II Row IV: $\leq 3.7\times 10^{-15}$;
- the $G(z)$ term in the form (3.4), p^+ integral done, = (B) (+ the second term of (3.5) in Row III(2)): $\leq 3.4\times 10^{-15}$;
- Group II Row IV: (i) + (ii) = 0: 5.5×10^{-16};
- the Fourier transforms: $q\int_0^\infty J_1(qr)\,e^{-ar}dr$ at $a = 10^{-2}, 3\times 10^{-3}, 10^{-3}$ (must tend to 1): $q = 0.7$: 0.9857, 0.9957, 0.9986; $q = 1.9$: 0.9947, 0.9984, 0.9995. $2\pi\int_0^\infty rJ_0(qr)\log r^2\,e^{-ar^2}dr\big/(-\frac{4\pi}{q^2})$ at $a = 4, 2, 1\times 10^{-3}$, extrapolated to $a = 0$ (must be 1): $q = 0.7$: 1.0350, 1.0169, 1.0083 $\to$ 0.99971; $q = 1.9$: 1.0045, 1.0022, 1.0011 $\to$ 1.00000.

4 Group I (with c.c.)

4.1 Row I

Only four ρ's: Sec. 7 (with the same coefficient for both logarithms).

4.2 Row II

Row II, first part. Your bracket $\left[\frac{11}{3} - 2\log\frac{k^+}{\Lambda} + \log(1 - \frac{k^+}{\vee})\right]$ multiplies the pole bracket; only its $-2\log\frac{k^+}{\Lambda}$ is a large logarithm. The term $\log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+}$ is not a BFKL term: its coefficient depends on $\vee$, it is bounded for every $\vee > k^+$ ($0 \leq -\log(1-\frac{k^+}{\vee})\log\frac{\vee}{k^+} \leq \log^2 2$, maximum at $\vee = 2k^+$) and vanishes for $\vee\to\infty$; it stays with the remaining terms. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = {}& \frac{1}{(2\pi)^3}\frac{g^4\,N_c}{32\pi^4}\frac{1}{k^+}\int_{w,w',x,x'}, e^{-ik(w'-w)}\,\frac{(x'-w')\cdot(x-w)}{(x'-w')^2\,(x-w)^2}\left[\left[-\frac{2}{\epsilon} - 2\gamma - \log\left[\frac{(x-w)^2\,\mu^2_{\overline{MS}}}{4}\right]\right]\times\right.\\ & \left.\left(-2\,\log\left(\frac{k^+}{\Lambda}\right)\right)\right]\times\\ & \left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ab}(w)\,\rho^{b}(x) - U^{ab}(x)\,\rho^{b}(x)\right]\end{aligned} \tag{4.1}$$

Terms $\propto \log\frac{\vee}{k^+}$: none.

Row II, second part, d^3N_{2a}. Line 1 has $\left[\log\frac{\vee}{k^+} + \log\frac{k^+}{\Lambda}\right]$ and enters both lists. In lines 2 and 3 the z integral is already done (hence $\frac{1}{32\pi^4}$); the measure should read $\int_{x,y,x'}$. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{2\,a}}{d^3k} = {}& -\frac{1}{(2\pi)^3}\frac{i\,g^4\,f^{hbc}}{32\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.\\ & \left.-\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\right]\\ & -\frac{1}{(2\pi)^3}\frac{i\,g^4\,f^{abc}}{32\pi^4}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\log\frac{(w-y)^2}{(x-y)^2}\right.\\ & \left.-\frac{(y-w)^i}{(y-w)^2}\log\frac{(w-x)^2}{(x-y)^2}\right]\times\left[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\right]\times\left[U^{bd}(x)\,U^{ce}(y)\,\{\rho^d(x),\rho^e(y)\}\right]\end{aligned} \tag{4.2}$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\frac{d^3N_{2\,a}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\frac{(w-z)\cdot(y-z)}{(w-z)^2\,(y-z)^2}\right.$$
$$\left. -\,\frac{(y-w)^i}{(y-w)^2}\,\frac{(w-z)\cdot(x-z)}{(w-z)^2\,(x-z)^2}\right]\times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big]$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{i\,g^4\,f^{hbc}}{32\pi^4}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{w,w'} e^{-ik\cdot(w'-w)}\int_{x,y,x',z}\frac{(x'-w')^i}{(x'-w')^2}\left[\frac{(x-w)^i}{(x-w)^2}\,\log\frac{(w-y)^2}{(x-y)^2} - \frac{(y-w)^i}{(y-w)^2}\,\log\frac{(w-x)^2}{(x-y)^2}\right]\times$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{ah}(w)\,\{\rho^b(x),\rho^c(y)\}\Big] \tag{4.3}$$

Row II, second part, d^3N_{2b}. No large logarithm (R^i depends on $\vee$ only through $\frac{k^+}{\vee}$). Terms $\propto \log \frac{k^+}{\Lambda}$: none. Terms $\propto \log \frac{\vee}{k^+}$: none.

4.3 Row III

Row III, d^3N_1. Lines 1 and 4 (your green $\frac{1}{\Lambda}$ brackets) are not logarithms. Together with line 1 of d^3N_2 and line 1 of d^3N_{3a} they cancel exactly (`Remaining_RC_DGLAP_finite.pdf`). Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left\{\right.$$
$$-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-z)^2+(x-w)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\left(-\log\left(\frac{k^+}{\Lambda}\right)\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-z)^2+(x-w)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\left.\right\} \tag{4.4}$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left\{\right.$$
$$-\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-z)^2+(x-w)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{\big[(x-w)^2+(x-z)^2\big]}{\big[(x-z)^2-(x-w)^2\big]^2}\,\times\log\Big(\frac{\vee}{k^+}\Big)$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$+\,\frac{ig^4\,f^{abd}}{16\pi^5}\,\frac{1}{k^+}\,\log\Big(\frac{\vee}{k^+}\Big)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$
$$-\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(x-z)\cdot(y-z)}{(x'-w')^2\,(x-w)^2\,(x-z)^2\,(y-z)^2}\,\log\Big(\frac{\vee}{k^+}\Big)$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]\left.\right\} \tag{4.5}$$

Row III, d^3N_2. Line 1 is a $\frac{1}{\Lambda}$ term (see d^3N_1). Your two red lines cancel each other and are left out. Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = -\,\frac{1}{(2\pi)^3}\int_{w,w'} e^{-ik(w'-w)}\left\{\right.$$
$$+\,\frac{ig^4\,f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,\big[(x-w)^2-(x-z)^2\big]^2}$$
$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^e(x)\,\rho^c(y) - U^{de}(x)\,U^{bc}(y)\,\rho^e(x)\,\rho^c(y)\Big]$$

$$
\begin{aligned}
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{4.6}
$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = &-\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Bigg\{\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,(x-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(x-w)\,(y-z)\cdot(z-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(z-w)\,(y-z)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]\,(x-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(x-z)\,(y-z)\cdot(z-w)}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x'}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-z)\,(x-w)^2}{(x'-w')^2\,(y-z)^2\,(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\}
\end{aligned}
\tag{4.7}
$$

Row III, d^3N_{3a}**.** Red: the first logarithmic line, with the edge kernel $\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}$, is not integrable at $z\to w$. Blue: it is replaced by $\mathcal{D}$ (fix F3, `BFKL_final.pdf` Sec. 2.1): $U^{bc}(z)\to U^{bc}(w)$ and the kernel $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)$. Without this change the three-ρ terms do not cancel (Sec. 8). Terms $\propto \log\frac{k^+}{\Lambda}$:

$$
\begin{aligned}
\frac{d^3N_{3a\,\mathrm{NLO}}}{d^3k} = &\frac{1}{(2\pi)^3}\int_{w,w'}e^{-ik(w'-w)}\Bigg\{\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-w)\,(x-z)\cdot(w-z)}{(x'-w')^2\,(y-w)^2\,(x-z)^2\,(w-z)^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(w)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&- \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\\
&+ \frac{ig^4 f^{abd}}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x'}\frac{(x'-w')\cdot(y-z)\,(z-w)\cdot(x-w)\,(x-z)^2}{(x'-w')^2(y-z)^2(z-w)^2\,[(x-w)^2-(x-z)^2]^2}\\
&\Big[U^{ab'}(x')\,\rho^{b'}(x')-U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y)-U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]
\end{aligned}
$$

$$\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big] \times \Big[U^{de}(x)\,U^{bc}(z)\,\rho^{e}(x)\,\rho^{c}(y) - U^{de}(x)\,U^{bc}(y)\,\rho^{e}(x)\,\rho^{c}(y)\Big]\Bigg\} \tag{4.8}$$

Terms $\propto \log \frac{\vee}{k^+}$: none.

Row III, d^3N_{3b}**.** No large logarithm. ($\mathcal{A}$ and the logarithms of d^3N_{3a}, d^3N_{3b} have finite limits for $\Lambda \to 0$; `Remaining_RC_DGLAP_finite.pdf`.) Terms $\propto \log \frac{k^+}{\Lambda}$: none.
Terms $\propto \log \frac{\vee}{k^+}$: none.

4.4 Row IV

Row IV. Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{ \\ + \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{k^+}{\Lambda}\right)\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\ \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]\end{aligned} \tag{4.9}$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik\cdot(w'-w)}\Bigg\{ \\ \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}\right. \\ \left. - \frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2} - \frac{(x-z)^j\,(x-w)^i}{(x-z)^2\,(x-w)^2} + \frac{\delta^{ij}}{2\,(z-w)^2}\right] \\ - \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\times\log\left(\frac{\vee}{k^+}\right) \\ + \frac{ig^4}{8\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\times \\ \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\Big[f^{cdc'}\,U^{ac}(w)\,U^{bd}(z)\,U^{be}(y)\,\rho^{e}(y)\rho^{c'}(x) - f^{abc}\,U^{be}(y)\,U^{cd}(x)\,\rho^{e}(y)\rho^{d}(x)\Big]\end{aligned} \tag{4.10}$$

4.5 Row V

Row V, first part. Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{1\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-z)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')\cdot(y-z)\,(z-x)\cdot(y'-x)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\ \Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\Big] \\ - \frac{1}{(2\pi)^3}\int_{x,z,y,x',y'w'} e^{-ik\cdot(w'-x)}\,\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')\cdot(y'-x)\,(z-x)\cdot(y-z)}{(x'-w')^2\,(z-x)^2\,(y-z)^2\,(y'-x)^2} \\ \times\Big[U^{ab'}(x')\,\rho^{b'}(x') - U^{ab'}(w')\,\rho^{b'}(x')\Big]\times\Big[f^{adc}\,U^{db}(x)\,U^{ce}(z)\,\{\rho^{e}(y),\rho^{b}(y')\} - f^{abc}\,U^{cd}(y)\,U^{be}(y')\,\{\rho^{d}(y),\rho^{e}(y')\}\Big]\end{aligned} \tag{4.11}$$

Terms $\propto \log \frac{\vee}{k^+}$: none.

Row V, second part, d^3N_{2a}**.** Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{2a\,\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-z)^2}\left[\frac{(y-z)^i\,(x-z)^m}{(x-z)^2}\right. \\ \left. + \frac{(y-x)^m\,(y-z)^i}{2\,(y-x)^2}\right]\times\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(x)\Big]\,\rho^{b'}(x')\,\rho^{f}(y) \\ - \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\,\frac{(x'-w')^i\,(z-x)^m}{(x'-w')^2\,(z-x)^2\,(y-x)^2} \\ \left[\frac{(y-x)^i\,(x-z)^m}{(x-z)^2} - \frac{(y-x)^i\,(y-z)^m}{2\,(y-z)^2}\right]\Big[U^{ab'}(x') - U^{ab'}(w')\Big]\Big[-f^{adc}\,f^{fbe}\,U^{db}(x)\,U^{ce}(z) + N_c\,U^{af}(x)\Big]\,\rho^{b'}(x')\,\rho^{f}(y)\end{aligned} \tag{4.12}$$

Terms $\propto \log \frac{\vee}{k^+}$: none.

Row V, second part, $d^3N_{2b\,\mathrm{UV}}$. Your bracket $\left[2\log\frac{k^+}{\Lambda}+2\log(1-\frac{\Lambda}{k^+})-\frac{11}{6}\right]$ is correct (E1 is now in your file; Row V has $0<p^+<k^+$, so no $\vee$). The poles of the $\log\frac{k^+}{\Lambda}$ terms cancel: Group I Row II : Row V 2b UV : Group II Row IV $= 1:-2:1$, sum 8.2×10^{-16} (relative; `ll_checks.py poles`). Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\frac{d^3N_{2b\,\mathrm{UV\,NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4N_c}{16\pi^4}\,\frac{1}{k^+}\,2\,\log\left(\frac{k^+}{\Lambda}\right)\int_{y,z,x',w'} e^{-ik\cdot(w'-z)} \left[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2_{\overline{MS}}}{4}\right]\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[U^{af}(y)-U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y). \tag{4.13}$$

Terms $\propto \log\frac{\vee}{k^+}$: none.

Row V, second part, $d^3N_{2b\,\mathrm{finite}}$. Lines 4–6 carry $\log\frac{k^+-\Lambda}{\Lambda}=\log\frac{k^+}{\Lambda}+\log(1-\frac{\Lambda}{k^+})$. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{2b\,\mathrm{finite\,NLO}}}{d^3k} =& -\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-x)}\left[\frac{(y-x)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right.\\
&\left.-\frac{(y-x)\cdot(x'-w')\,(y-z)\cdot(z-x)}{2(y-z)^2\,(x'-w')^2\,(z-x)^2\,(y-x)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\left[\frac{(y-z)\cdot(x'-w')\,(x-z)\cdot(z-x)}{(x-z)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right.\\
&\left.+\frac{(y-x)\cdot(z-x)\,(y-z)\cdot(x'-w')}{2(y-x)^2\,(x'-w')^2\,(z-x)^2\,(y-z)^2}\right]\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(x)\Big]\rho^{b'}(x')\,\rho^f(y)\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\int_{x,y,z,x',w'} e^{-ik\cdot(w'-z)}\,2\,\log\left(\frac{k^+}{\Lambda}\right)\\
&\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2}\,\frac{(y-x)\cdot(y-z)}{(y-x)^2\,(x-z)^2}\times\Big[U^{ab'}(x')-U^{ab'}(w')\Big]\Big[N_c\,U^{af}(y)-N_c\,U^{af}(z)\Big]\rho^{b'}(x')\,\rho^f(y)\end{aligned} \tag{4.14}$$

Terms $\propto \log\frac{\vee}{k^+}$: none.

5 Group II (no c.c.)

5.1 Row I

Row I, first part, first line. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} =& \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{\\
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\\
&\Big[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]\end{aligned} \tag{5.1}$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\begin{aligned}\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} =& \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{\\
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\,\log\left(\frac{\vee}{k^+}\right)\times\\
&\left[\frac{\delta^{ij}}{2\,(z-w)^2}+\frac{(x-z)^j\,(z-w)^i}{(z-w)^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w)^i}{(x-w)^2\,(x-z)^2}-\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}\right]\\
&+\frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2\,(y-z)^2\,(z-w)^2}\\
&-\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i\,(y-z)^j}{(x'-w')^2\,(y-z)^2}\left[\frac{(x-w)^i\,(z-w)^j}{(z-w)^2\,(x-w)^2}+\frac{(x-z)^j\,(x-w)^i}{2\,(x-z)^2\,(x-w)^2}\right]\Bigg\}\\
&\Big[U^{ac'}(w')\,\{\rho^{c'}(x'),\rho^b(y)\}-U^{bc'}(z)\,U^{ad'}(x')\,U^{c'e'}(y)\,\{\rho^{d'}(x'),\rho^{e'}(y)\}\Big]\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]\end{aligned} \tag{5.2}$$

Row I, first part, second line. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\frac{d^3N}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'} e^{-ik(w'-w)}\Bigg\{$$

$$+\ \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}$$
$$\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]\quad(5.3)$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\frac{d^3N}{d^3k}=\frac{1}{(2\pi)^3}\int_{x,y,z,w,x',w'}e^{-ik(w'-w)}\Bigg\{$$
$$\frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\log\left(\frac{\vee}{k^+}\right)\times$$
$$\left[\frac{\delta^{ij}}{2\,(z-w')^2}+\frac{(x-z)^j\,(z-w')^i}{(z-w')^2\,(x-z)^2}-\frac{(x-z)^j\,(x-w')^i}{(x-w')^2\,(x-z)^2}-\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}\right]$$
$$-\ \frac{ig^4}{32\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w)\cdot(y-z)}{(x'-w)^2\,(y-z)^2\,(z-w')^2}$$
$$+\ \frac{ig^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w)^i\,(y-z)^j}{(x'-w)^2\,(y-z)^2}\,\left[\frac{(x-w')^i\,(z-w')^j}{(z-w')^2\,(x-w')^2}+\frac{(x-z)^j\,(x-w')^i}{2\,(x-z)^2\,(x-w')^2}\right]\Bigg\}$$
$$\times\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x)-f^{ac'd'}\,U^{bc'}(z)\,U^{d'e'}(x)\,\rho^{e'}(x)\right]\times\left[U^{ac}(w)\,\{\rho^c(x'),\rho^b(y)\}-U^{bc}(z)\,U^{ad}(x')\,U^{ce}(y)\,\{\rho^d(x'),\rho^e(y)\}\right]\quad(5.4)$$

Row I, second part, d^3N_{2a}. No large logarithm: $\vee$ cancels between numerator and denominator of the first logarithm, and the second has the finite limit $\log\frac{(x-w)^2}{(x'-w')^2}$. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$: none.

Row I, second part, d^3N_{2b}. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)$$
$$-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Big]\quad(5.5)$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\frac{d^3N_{2b\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\left[(x-z)^2-\frac{(x-w)^2\,(x'-z)^2}{(x'-w')^2}\right]\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)$$
$$-\,f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^d(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^e(x)\Big]$$
$$-\,\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-z)^2}{[(x'-w')^2\,(x-z)^2\,(x-w)^2-(x-w)^4\,(x'-z)^2]}\times$$
$$\left[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2}+\frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2}\right.$$
$$\left.+\,\frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2}+\frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\right]\times\log\left(\frac{\vee}{k^+}\right)\times$$
$$\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^d(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^e(x)$$

$$
\begin{aligned}
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-z)^2}{[(x'-w')^4\,(x-z)^2 \,-(x-w)^2\,(x'-z)^2\,(x'-w')^2]} \times \\
&\Bigg[\frac{(x'-w')\cdot(x-w)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} + \frac{(x'-w')\cdot(x-w)\,(x'-z)\cdot(z-w)}{2\,(z-w)^2\,(x'-z)^2} \\
&+ \frac{(x-w)\cdot(x'-w')\,(x-z)\cdot(z-w')}{2\,(x-z)^2\,(z-w')^2} + \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-z)^2\,(x'-z)^2}\Bigg] \times \log\Big(\frac{\vee}{k^+}\Big) \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{5.6}
$$

Row I, second part, d^3N_{2c}. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N_{2c\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x'-w')^2}{(x'-z)^2}\, \frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\Bigg[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\Bigg] \times \log\Big(\frac{\vee}{k^+}\Big) \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&- \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-w)^2}{(x-z)^2}\, \frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\Bigg[\frac{(x'-z)\cdot(x-z)\,(z-w)\cdot(z-w')}{(z-w)^2\,(z-w')^2} - \frac{(x'-z)\cdot(x-z)\,(x'-w')\cdot(z-w)}{2\,(z-w)^2\,(x'-w')^2} - \frac{(x-w)\cdot(z-w')\,(x-z)\cdot(x'-z)}{2\,(x-w)^2\,(z-w')^2} \\
&+ \frac{(x-z)\cdot(x'-z)\,(x'-w')\cdot(x-w)}{4\,(x-w)^2\,(x'-w')^2}\Bigg] \times \log\Big(\frac{\vee}{k^+}\Big) \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big]
\end{aligned}
\tag{5.7}
$$

Row I, second part, d^3N_{2d}. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N_{2d\,\mathrm{NLO}}}{d^3k} &= -\frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+}\, \log\Big(\frac{\vee}{k^+}\Big) \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{1}{[(x-z)^2-(x-w)^2]}\, \frac{1}{[(x'-w')^2-(x'-z)^2]} \\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2} \\
&+ \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg] \\
&\Big[f^{c'bd'}\, f^{cbd}\, U^{ac'}(w')\, U^{ac}(w)\, \rho^{d'}(x')\rho^{d}(x) - f^{c'bd'}\, f^{acd}\, U^{ac'}(w')\, U^{bc}(z)\, U^{de}(x)\, \rho^{d'}(x')\rho^{e}(x) \\
&- f^{ac'd'}\, f^{cbd}\, U^{ac}(w)\, U^{bc'}(z)\, U^{e'd'}(x')\, \rho^{e'}(x')\, \rho^{d}(x) + N_c\, U^{e'd}(x')\, U^{de}(x)\, \rho^{e'}(x')\, \rho^{e}(x)\Big] \\
&+ \frac{1}{(2\pi)^3}\, \frac{g^4}{8\pi^5}\, \frac{1}{k^+} \int_{z,w,w',x,x'} e^{-ik(w'-w)}\, \frac{(x-z)^2}{[(x-z)^2-(x-w)^2]}\, \frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]} \\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2} \\
&+ \frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2} + \frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg] \times
\end{aligned}
$$

$$
\begin{aligned}
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-z)^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-w)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-z)^2\,(z-w')^2}\\
&+\frac{(x'-w')\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}+\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-z)^2}\Bigg]\times\\
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{5.8}
$$

Row I, second part, d^3N_{2e}. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$:

$$
\begin{aligned}
\frac{d^3N_{2e\,\mathrm{NLO}}}{d^3k}&=\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{1}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2-(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x-w)^2}{[(x-z)^2-(x-w)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times\\
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'}e^{-ik(w'-w)}\,\frac{(x'-w')^2}{[(x'-w')^2-(x'-z)^2]}\,\frac{1}{[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2]}\\
&\Bigg[\frac{(x-z)\cdot(z-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x-z)\cdot(x-w)\,\left[(x'-z)^2-(x'-w')^2\right]}{4\,(x-w)^2\,(z-w')^2}\\
&+\frac{(x'-z)\cdot(z-w')\,\left[(x-z)^2-(x-w)^2\right]}{2\,(z-w)^2\,(z-w')^2}-\frac{(x'-z)\cdot(x'-w')\,\left[(x-z)^2-(x-w)^2\right]}{4\,(z-w)^2\,(x'-w')^2}\Bigg]\times\\
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Bigg[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Bigg]
\end{aligned}
\tag{5.9}
$$

Row I, second part, d^3N_{2f}. Terms $\propto \log\frac{k^+}{\Lambda}$: none.
Terms $\propto \log\frac{\vee}{k^+}$:

$$\begin{aligned}
\frac{d^3N_{2f\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\log\left(\frac{\vee}{k^+}\right)\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{\left[(x'-w')^2\,(x-w)^2-(x-z)^2\,(x'-z)^2\right]}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]\,\left[(x'-z)^2-(x'-w')^2\right]}\\
&\left[f^{c'bd'}\,U^{ac'}(w')\,\rho^{d'}(x')-f^{ac'd'}\,U^{bc'}(z)U^{e'd'}(x')\,\rho^{e'}(x')\right]\times\left[f^{cbd}\,U^{ac}(w)\,\rho^{d}(x)-f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^{e}(x)\right]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x-z)^2\,(x-w)^2}{(z-w)^2\,(z-w')^2\,\left[(x-z)^2-(x-w)^2\right]}\,\frac{\left[(x'-z)^2-(x'-w')^2\right]}{\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\\
&\log\left(\frac{\vee}{k^+}\right)\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4\,d_\perp}{32\pi^5}\,\frac{1}{k^+}\int_{z,w,w',x,x'} e^{-ik(w'-w)}\,\frac{(x'-w')^2\,(x'-z)^2}{(z-w)^2\,(z-w')^2}\\
&\frac{\left[(x-z)^2-(x-w)^2\right]}{\left[(x'-z)^2-(x'-w')^2\right]\,\left[(x'-w')^2\,(x-z)^2-(x-w)^2\,(x'-z)^2\right]}\times\log\left(\frac{\vee}{k^+}\right)\times\\
&\Big[f^{c'bd'}\,f^{cbd}\,U^{ac'}(w')\,U^{ac}(w)\,\rho^{d'}(x')\rho^{d}(x)-f^{c'bd'}\,f^{acd}\,U^{ac'}(w')\,U^{bc}(z)\,U^{de}(x)\,\rho^{d'}(x')\rho^{e}(x)\\
&-f^{ac'd'}\,f^{cbd}\,U^{ac}(w)\,U^{bc'}(z)\,U^{e'd'}(x')\,\rho^{e'}(x')\,\rho^{d}(x)+N_c\,U^{e'd}(x')\,U^{de}(x)\,\rho^{e'}(x')\,\rho^{e}(x)\Big]
\end{aligned}\tag{5.10}$$

5.2 Row II

Only four ρ's: Sec. 7 (with the same coefficient for both logarithms).

5.3 Row III

Only four ρ's: Sec. 7 (with the same coefficient for both logarithms).

5.4 Row IV

Row IV. BFKL terms in line 1. Line 1 is a ξ integral down to $\xi=\frac{k^+}{\vee}$; it contains $\log\frac{\vee}{k^+}$ terms that are brought out by exact identities (Sec. 3):

(i) line 2: $\log(\frac{\vee}{k^+}-1)=\log\frac{\vee}{k^+}+\log(1-\frac{k^+}{\vee})$; its $\log\frac{\vee}{k^+}$ part is the first line below;

(ii) line 1, the subtraction in $\frac{\xi}{(1-\xi)_+}$: $-g(1)\log\frac{\vee}{k^+}$, Eq. (3.3), exactly minus (i), poles included (second line, blue);

(iii) line 1, the $\log(y'-y)^2$ term of $\frac{1-\xi}{\xi}$ at $y'=y$: $\log\frac{\vee}{k^+}H_4$, Eq. (3.6) (third line, blue).

(i) and (ii) cancel: the BFKL term of this row is $\log\frac{\vee}{k^+}H_4$. Its $\log\frac{k^+}{\Lambda}$ term is as you have it. Terms $\propto\log\frac{k^+}{\Lambda}$:

$$\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
\end{aligned}\tag{5.11}$$

Terms $\propto\log\frac{\vee}{k^+}$:

$$\begin{aligned}
\frac{d^3N_{\mathrm{NLO}}}{d^3k} &= +\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
&-\frac{g^2}{4\pi^2}\,\frac{1}{(2\pi)^3}\,\frac{g^2}{2\pi^2}\,\frac{N_c}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,y,x',y'} e^{-ik\cdot(y'-y)}\,\frac{(x-y)\cdot(x'-y')}{(x-y)^2\,(x'-y')^2}\left[-\frac{d_\perp}{\epsilon}-d_\perp\,\gamma_E\right.\\
&\left.-\frac{d_\perp}{2}\,\log\left(\frac{(y'-y)^2\,\mu^2_{\overline{MS}}}{4}\right)-1\right]\left[U^{e'c}(y')-U^{e'c}(x')\right]\times\left[U^{ec}(y)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)\\
&\frac{1}{(2\pi)^3}\,\frac{g^4\,N_c}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\,\frac{(w-z)\cdot(w'-z)}{(w-z)^2\,(w'-z)^2}\,\frac{(x-z)\cdot(x'-z)}{(x-z)^2\,(x'-z)^2}\\
&\left[U^{e'c}(z)-U^{e'c}(x')\right]\times\left[U^{ec}(z)-U^{ce}(x)\right]\,\rho^{e'}(x')\rho^{e}(x)
\end{aligned}\tag{5.12}$$

6 Group III (with c.c.)

6.1 Row I

Row I. Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times \\ \Big[U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')\rho^{c'}(x')\rho^{e'}(y)\Big]\times \\ \Big[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\Big]\end{aligned} \tag{6.1}$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right. \\ \left.-\frac{(x-z)^j(x-w)^i}{(x-w)^2(x-z)^2}-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}\right] \\ +\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times \\ \Big[U^{ab'}(x')U^{c'd'}(y)U^{c'b}(z)\rho^{b'}(x')\rho^{d'}(y)-U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')\rho^{c'}(x')\rho^{e'}(y)\Big]\times \\ \Big[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\Big]\end{aligned} \tag{6.2}$$

6.2 Row II

Row II. Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times \\ \Big[U^{c'd'}(y)U^{ab'}(x')U^{bc'}(z)\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\rho^b(y)\rho^{b'}(x')\Big]\times\Big[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\Big]\end{aligned} \tag{6.3}$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\begin{aligned}\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\int_{x,y,x',z,w,w'} e^{-ik(w'-w)}\Bigg\{ \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\times\log\left(\frac{\vee}{k^+}\right)\times\left[\frac{\delta^{ij}}{2(z-w)^2}+\frac{(x-z)^j(z-w)^i}{(z-w)^2(x-z)^2}\right. \\ \left.-\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}-\frac{(x-z)^j(x-w)^i}{(x-z)^2(x-w)^2}\right] \\ +\frac{ig^4}{16\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')\cdot(y-z)}{(x'-w')^2(y-z)^2(z-w)^2} \\ -\frac{ig^4}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\frac{(x'-w')^i(y-z)^j}{(x'-w')^2(y-z)^2}\left[\frac{(x-w)^i(z-w)^j}{(z-w)^2(x-w)^2}+\frac{(x-z)^j(x-w)^i}{2(x-z)^2(x-w)^2}\right]\Bigg\}\times \\ \Big[U^{c'd'}(y)U^{ab'}(x')U^{bc'}(z)\rho^{d'}(y)\rho^{b'}(x')-U^{ab'}(x')\rho^b(y)\rho^{b'}(x')\Big]\times\Big[f^{cbd}U^{ac}(w)\rho^d(x)-f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\Big]\end{aligned} \tag{6.4}$$

6.3 Row III

Row III, first part. BFKL terms in line 1. Line 1 is a p^+ integral up to $\vee - k^+$; exact identities bring out (Sec. 3):

(i) line 2: the $\log \frac{\vee}{k^+}$ part of $\log(\frac{\vee}{k^+} - 1)$ (first line below);

(ii) line 1, the subtraction of $\left[\frac{1}{p^+}\right]_+$: $-f(0) \log \frac{\vee}{k^+}$, Eq. (3.2), exactly minus (i) (blue);

(iii) line 1, the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term with $G(y') \to G(z)$: the term (B), Eq. (3.4) (blue).

(i) and (ii) cancel, so the BFKL term of this part is (B). Its $\log \frac{k^+}{\Lambda}$ term is as you have it. Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\frac{d^3 N_{1\,\mathrm{NLO}}}{d^3 k} = -\frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big] \times \Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big] \tag{6.5}$$

Terms $\propto \log \frac{\vee}{k^+}$:

$$\begin{aligned}
\frac{d^3 N_{1\,\mathrm{NLO}}}{d^3 k} =& -\frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big] \times \Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big]\\
&+ \frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big] \times \Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big]\\
&- \frac{1}{(2\pi)^3}\,\frac{i g^4\, f^{c'd'a}}{16\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',w',z,y,x} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}\\
&\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&\times \Big[U^{ac}(w)\,\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\{\rho^d(y),\rho^e(x)\}\Big]
\end{aligned} \tag{6.6}$$

Row III, second part. (i), (ii), (iii) as in the first part; (B) comes from the $-\frac{p^+}{k^{+2}}$ term, Eqs. (3.4), (3.5). **C5**: the measure is $\int_{w,x',y',z,x}$ (the row has $\rho(x')\rho(x)$; there is no y and no w'). Terms $\propto \log \frac{k^+}{\Lambda}$:

$$\begin{aligned}
\frac{d^3 N_{2\,\mathrm{NLO}}}{d^3 k} =& -\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{k^+}{\Lambda}\right)\int_{w,y',z,y,x}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\Big[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]\\
&+\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x}\int_{w,x',y',z,x} e^{-ik\cdot(y'-w)}\left[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right.\\
&\left.+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2\,(x-z)^2\,(x-w)^2\,(y'-z)^2\,(x'-y')^2}\right]\Big[f^{c'd'a}\,U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\\
&\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]\\
&-\frac{1}{(2\pi)^3}\,\frac{g^4}{8\pi^5}\,\frac{1}{k^+}\,\log\left(\frac{\vee}{k^+}\right)\int_{w,x',w',z,x} e^{-ik\cdot(w'-w)}\,\frac{(x'-z)\cdot(x-z)}{(x'-z)^2\,(x-z)^2}\\
&\times\left[\frac{(w'-z)\cdot(z-w)}{(w'-z)^2\,(z-w)^2} - \frac{(w'-z)\cdot(x-w)}{2\,(w'-z)^2\,(x-w)^2}\right]\\
&\Big[f^{c'd'a}\,U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - f^{c'd'a}\,U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\\
&\times\Big[f^{cbd}\,U^{ac}(w)\,\rho^d(x) - f^{acd}\,U^{bc}(z)\,U^{de}(x)\,\rho^e(x)\Big]
\end{aligned} \tag{6.8}$$

6.4 Row IV

Only four ρ's: Sec. 7 (with the same coefficient for both logarithms).

6.5 Row V

Row V. (i), (ii), (iii) as in Row III: (i) and (ii) cancel, and (B) comes from the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\Big] \tag{6.9}$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\Big]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\Big]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'ca}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,\rho^{e'}(x') - U^{c'e'}(x')\,\rho^{e'}(x')\Big]\Big[U^{ce}(x)\,U^{ad}(y)\,\rho^{e}(x)\rho^{d}(y) - U^{ad}(w)\,U^{ce}(x)\,\rho^{e}(x)\rho^{d}(y)\Big] \tag{6.10}$$

6.6 Row VI

Row VI. (i), (ii), (iii) as in Row III: (i) and (ii) cancel, and (B) comes from the $-\frac{1}{k^+}\delta_{im}\delta_{jk'}$ term. Terms $\propto \log\frac{k^+}{\Lambda}$:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{k^+}{\Lambda}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big] \tag{6.11}$$

Terms $\propto \log\frac{\vee}{k^+}$:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big]$$
$$+\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\,\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2\,(x'-y')^2\,(y-w)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(y')\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big]$$
$$-\frac{1}{(2\pi)^3}\frac{ig^4\,f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\log\left(\frac{\vee}{k^+}\right)\int_{w,w',z,y,x,x'} e^{-ik\cdot(w'-w)}\,\frac{(w'-z)\cdot(y-w)\,(x'-z)\cdot(x-z)}{(w'-z)^2\,(y-w)^2\,(x'-z)^2\,(x-z)^2}$$
$$\Big[U^{e'c'}(z)\,U^{d'b}(z)\,\rho^{e'}(x') - U^{bd'}(z)\,U^{c'e'}(x')\,\rho^{e'}(x')\Big]$$
$$\times\Big[U^{bc}(z)\,U^{ad}(y)\,U^{ce}(x)\,\rho^{d}(y)\,\rho^{e}(x) - U^{ad}(y)\,\rho^{d}(y)\,\rho^{b}(x)\Big] \tag{6.12}$$

7 Four ρ's

Group I Row I, Group II Rows II and III and Group III Row IV have only four ρ's: your `4rho.tex`. Every four-ρ term carries $\left[\log\frac{k^+}{\Lambda}+\log\frac{\vee}{k^+}\right]$: the same coefficient for both logarithms.

- The four-ρ terms cancel after symmetrization (Sec. 2: 1.3×10^{-16}).
- Their three-ρ remainders, your final forms, enter both three-ρ lists of Sec. 8:

$$\frac{d^3N_{\rm NLO}}{d^3k} = -\frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$
$$\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{d}(x')\rho^{a}(y)\rho^{c}(x)\,U^{de}(w')U^{eb}(w)$$
$$-\mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{e}(x')\rho^{a}(y)\rho^{c}(x)\,U^{db}(w)U^{ed}(x')$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{c}(x')\rho^{a}(y)\rho^{e}(x)\,U^{db}(w')U^{ed}(x)$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\,f^{abc}\rho^{c}(x')\rho^{a}(y)\rho^{e}(x)U^{bd}(w')U^{ed}(w)\Big] \tag{7.1}$$

$$\frac{d^3N_{\rm NLO}}{d^3k} = \frac{i}{(2\pi)^3}\frac{g^4}{8\pi^3}\frac{1}{2\pi^2}\frac{1}{k^+}\left[\log\left[\frac{k^+}{\Lambda}\right]+\log\left[\frac{\vee}{k^+}\right]\right]\int_{x,y,z,x',w,w'} e^{-ik(w'-w)}\Big[$$

$$
\begin{aligned}
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^d(x')\rho^h(y)\rho^e(x)U^{db}(w')U^{ec}(x)U^{ah}(y)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{eb}(w)U^{ah}(y)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x'-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^d(x')\rho^e(x)\rho^h(y)U^{dc}(x')U^{be}(x)U^{ah}(y)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(x-w)\,\mathcal{K}(x-z)\cdot\mathcal{K}(y-z)\, f^{abc}\rho^e(x')\rho^h(y)\rho^d(x)U^{dc}(x)U^{be}(x')U^{ah}(y)\Big]
\end{aligned}
\tag{7.2}
$$

- Their two-ρ remainders (your "Two ρ contribution", with the log bracket restored in the first part and F1, F2) enter both two-ρ sums, Secs. 9 and 10.

8 The three-ρ terms cancel

All terms with three ρ's of Secs. 4–7, for each logarithm separately. Which terms have three ρ's is decided numerically ($\rho \to 2\rho$ multiplies them by 8).

Method.

- The charges are integration variables, so only the totally symmetric part of $\rho\rho\rho$ can cancel against other terms. The rest (commutators, $[\rho^a(u),\rho^b(v)] = if^{abc}\rho^c(u)\delta(u-v)$) are two-$\rho$ terms; they are in Secs. 9 and 10. The symmetric part is evaluated with classical charges.
- In every term the integration variables are renamed so that the phase is $e^{-ik\cdot(w'-w)}$, the soft gluon is z and the charges are x, y, x': in Group I Row V part 1 the measured gluon of the amplitude is at z (line 3) or x (line 4); in Group III Rows III, V, VI and in (B) the conjugate one is at y'.
- Each term is summed over the 6 placements of its three charges and, for Groups I and III, its c.c. is added (conjugate, $w \leftrightarrow w'$). Group II Row I is self-conjugate as written: its second line is the c.c. of the first. The `4rho.tex` lists are self-conjugate as written.
- In Group I Row II d^3N_{2a}, lines 2 and 3, the z integral is done. For a pointwise test it is written back: $\log\frac{(w-y)^2}{(x-y)^2} = -\frac{1}{\pi}\int_z \big[K(w,y,z) - K(x,y,z)\big]$, $K(a,b,z) = \mathcal{K}(a-z)\cdot\mathcal{K}(b-z)$.
- Random points, random SU(3) adjoint Wilson lines with $U^{bc} = U^{cb}$, random charges; three configurations.

Values at one configuration, in units of $i\mathcal{N}\,e^{-ik\cdot(w'-w)}$, with the large logarithm set to 1 (all values are real in these units):

Three ρ's $\propto \log\frac{k^+}{\Lambda}$ (22 terms).

term		direct	c.c.	sum
Group I II(2a), line 1	Eq. (4.2)	−0.8549	−14.6946	−15.5494
Group I II(2a), line 3	Eq. (4.2)	+1.0769	+0.2120	+1.2889
Group I III d^3N_1, line 2	Eq. (4.4)	+45.4456	+0.0289	+45.4745
Group I III d^3N_1, line 5	Eq. (4.4)	−45.4456	−0.0289	−45.4745
Group I III d^3N_1, line 7	Eq. (4.4)	−0.4492	−1.2495	−1.6987
Group I III d^3N_2, line 5	Eq. (4.6)	−19.8506	+0.1921	−19.6585
Group I III d^3N_2, line 6	Eq. (4.6)	−30.9162	−2.0644	−32.9806
Group I III d^3N_{3a}, line 2 ($\mathcal{D}$)	Eq. (4.8)	−15.7868	−25.4825	−41.2693
Group I III d^3N_{3a}, line 3	Eq. (4.8)	+30.9162	+2.0644	+32.9806
Group I III d^3N_{3a}, line 4	Eq. (4.8)	+19.8506	−0.1921	+19.6585
Group I IV, line 3	Eq. (4.9)	−0.0717	−9.8576	−9.9294
Group I V(1), line 3	Eq. (4.11)	−0.2873	−0.4411	−0.7284
Group I V(1), line 4	Eq. (4.11)	−0.2873	−0.4411	−0.7284
Group II I(1) A, line 3	Eq. (5.1)	−1.8952	(self-conj.)	−1.8952
Group II I(1) B, line 3	Eq. (5.3)	+5.2217	(self-conj.)	+5.2217
Group III I, line 3	Eq. (6.1)	+0.0717	+9.8576	+9.9294
Group III II, line 3	Eq. (6.3)	+1.5672	+3.4600	+5.0271
Group III III(1), line 2	Eq. (6.5)	−0.1498	+0.5177	+0.3679
Group III V, line 2	Eq. (6.9)	+25.4825	+15.7868	+41.2693
Group III VI, line 2	Eq. (6.11)	+2.6061	+1.5813	+4.1874
`4rho.tex`, two Wilson lines	Eq. (7.1)	+6.3001	(in the term)	+6.3001
`4rho.tex`, three Wilson lines	Eq. (7.2)	−1.7932	(in the term)	−1.7932
total				$-3.5e-14$

Largest relative residual over three configurations: final: 7.7×10^{-16}; edge kernel (as written) instead of D: 5.9×10^{-1}.

Three ρ's $\propto \log \frac{\vee}{k^+}$ (35 terms).

term		direct	c.c.	sum
Group I II(2a), line 1	Eq. (4.3)	-0.8549	-14.6946	-15.5494
Group I II(2a), line 2	Eq. (4.3)	-0.5232	$+3.4724$	$+2.9492$
Group I III d^3N_1, line 2	Eq. (4.5)	$+45.4456$	$+0.0289$	$+45.4745$
Group I III d^3N_1, line 3	Eq. (4.5)	-45.4456	-0.0289	-45.4745
Group I III d^3N_1, line 7	Eq. (4.5)	-0.4492	-1.2495	-1.6987
Group I III d^3N_1, line 8	Eq. (4.5)	$+0.8984$	$+2.4989$	$+3.3973$
Group I III d^3N_2, line 2	Eq. (4.7)	$+1.5244$	$+1.5741$	$+3.0986$
Group I III d^3N_2, line 5	Eq. (4.7)	-19.8506	$+0.1921$	-19.6585
Group I III d^3N_2, line 6	Eq. (4.7)	-30.9162	-2.0644	-32.9806
Group I III d^3N_2, line 7	Eq. (4.7)	$+4.9673$	-2.0824	$+2.8850$
Group I III d^3N_2, line 8	Eq. (4.7)	$+3.0331$	-0.7494	$+2.2837$
Group I III d^3N_2, line 9	Eq. (4.7)	$+30.9162$	$+2.0644$	$+32.9806$
Group I III d^3N_2, line 10	Eq. (4.7)	$+9.5357$	$+0.0611$	$+9.5968$
Group I III d^3N_2, line 11	Eq. (4.7)	-21.2938	-0.2070	-21.5009
Group I III d^3N_2, line 12	Eq. (4.7)	$+19.8506$	-0.1921	$+19.6585$
Group I IV, line 1	Eq. (4.10)	$+3.4365$	$+1.9039$	$+5.3403$
Group I IV, line 2	Eq. (4.10)	-0.5655	-1.2722	-1.8377
Group I IV, line 3	Eq. (4.10)	-0.0717	-9.8576	-9.9294
Group II I(1) A, line 1	Eq. (5.2)	$+1.9040$	(self-conj.)	$+1.9040$
Group II I(1) A, line 2	Eq. (5.2)	$+1.5852$	(self-conj.)	$+1.5852$
Group II I(1) A, line 3	Eq. (5.2)	-1.8952	(self-conj.)	-1.8952
Group II I(1) B, line 1	Eq. (5.4)	-2.0102	(self-conj.)	-2.0102
Group II I(1) B, line 2	Eq. (5.4)	$+0.0433$	(self-conj.)	$+0.0433$
Group II I(1) B, line 3	Eq. (5.4)	$+5.2217$	(self-conj.)	$+5.2217$
Group III I, line 1	Eq. (6.2)	-3.4365	-1.9039	-5.3403
Group III I, line 2	Eq. (6.2)	$+0.5655$	$+1.2722$	$+1.8377$
Group III I, line 3	Eq. (6.2)	$+0.0717$	$+9.8576$	$+9.9294$
Group III II, line 1	Eq. (6.4)	-3.1195	-1.3034	-4.4229
Group III II, line 2	Eq. (6.4)	$+4.0897$	$+0.4887$	$+4.5784$
Group III II, line 3	Eq. (6.4)	$+1.5672$	$+3.4600$	$+5.0271$
(B), Group III Row III(1)	Eq. (6.6)	-1.4036	-2.2332	-3.6368
(B), Group III Row V	Eq. (6.10)	$+1.4036$	$+2.2332$	$+3.6368$
(B), Group III Row VI	Eq. (6.12)	-0.0000	$+0.0000$	-0.0000
`4rho.tex`, two Wilson lines	Eq. (7.1)	$+6.3001$	(in the term)	$+6.3001$
`4rho.tex`, three Wilson lines	Eq. (7.2)	-1.7932	(in the term)	-1.7932
total				$+4.9e-15$

Largest relative residual over three configurations: final: 1.5×10^{-16}; without (B): 3.1×10^{-16}.

So:

- **The three-ρ terms $\propto \log \frac{k^+}{\Lambda}$ cancel**, with $\mathcal{D}$ in place of the edge-kernel term of Group I Row III d^3N_{3a}. With the edge kernel they do not.
- **The three-ρ terms $\propto \log \frac{\vee}{k^+}$ cancel.** The (B) terms of Group III Rows III(1) and V cancel each other, that of Row VI vanishes by itself after symmetrization; so the three-ρ check does not test (B). (B) is tested by the two-ρ sum, Sec. 10.
- Adding a c.c. to the `4rho.tex` lists would double them: they already contain it.
- Group I Row IV and Group III Row I cancel each other line by line, and so do several pairs of lines in Group I Row III.

9 The two-ρ terms $\propto \log \frac{k^+}{\Lambda}$: JIMWLK

Number of terms (green / purple): genuine, expanded: 135 / 4; `4rho.tex` lists: 0 / 30; from reordering three and four ρ's: 208 / 127. No orange term appears.

9.1 Genuine two-ρ terms of the rows

These are the two-ρ terms $\propto \log \frac{k^+}{\Lambda}$ of Secs. 4–6 (Group I Rows II and V, Group II Rows I and IV, Group III Row III part 2), written in a common notation. To compare term by term, every dimensionally regulated bracket is written back as the z integral it comes from (exact in $d_\perp = 2-\epsilon$, $\overline{\text{MS}}$):

- Group I Row II: $\frac{2}{\epsilon} + 2\gamma + \log \frac{(x-w)^2\mu^2}{4} = -\frac{1}{\pi}\int_z \big[K(x,w,z) - K(w,w,z)\big]$;

- Group I Row V 2b UV: $\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y-z)^2\mu^2}{4}=\frac{1}{\pi}\int_x\big[K(z,z,x)-K(y,z,x)\big]$;
- Group II Row IV: $-\frac{d_\perp}{\epsilon}-d_\perp\gamma_E-\frac{d_\perp}{2}\log\frac{(y'-y)^2\mu^2}{4}-1=\frac{1}{\pi}\int_z\big[K(y,y',z)-\frac{1}{2}K(y,y,z)-\frac{1}{2}K(y',y',z)\big]+O(\epsilon)$.

The terms are expanded, the charges are named x (emission ending at w) and x' (ending at w'), and the colour algebra is done. Group I and Group III include their c.c.

- **Row V 2b: the UV term and the finite counterterm cancel each other.** Both have the kernel $\mathcal{K}(x'-w')\cdot\mathcal{K}(y-z)\,[K(z,z,x)-K(y,z,x)]$ and the colour $N_c[U(y)-U(z)]$.
  - UV: $+\frac{g^4N_c}{8\pi^4}\cdot\frac{1}{\pi}\int_x$.
  - Counterterm: $-\frac{g^4}{8\pi^5}N_c$, pointwise in x.

  So the $\log\frac{k^+}{\Lambda}$ part of Row V 2b is just its two endpoint terms. This is the same as before the UV subtraction.
- Row V 2b UV is taken with the coefficient $2\log\frac{k^+}{\Lambda}$ (E1).

Group I, Row II.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}II}=\mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\Bigg[&\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Bigg]\end{aligned}\tag{9.1}$$

Group I, Row V, part 2a.

$$\begin{aligned}\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2a}=\mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\Bigg[&\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\end{aligned}$$

$$
\begin{aligned}
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{9.2}
$$

Group I, Row V, part 2b finite: the two endpoint terms (B_2, B_4).

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\ \mathrm{end}} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w')\,U^{ac}(z)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w)\,U^{ac}(z)\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{9.3}
$$

Group I, Row V, part 2b UV (bracket written as $\frac{1}{\pi}\int_x[K(z,z,x)-K(y,z,x)]$).

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\ \mathrm{UV}} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[&\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

$$
\begin{aligned}
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{9.4}
$$

Group I, Row V, part 2b finite: the counterterm.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G1\text{-}V\ 2b\,\mathrm{CT}} &= \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{9.5}
$$

Group II, Row I.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G2\text{-}I} &= \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{9.6}
$$

Group II, Row IV.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G2\text{-}IV} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{9.7}
$$

Group III, Row III, second part.

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{G3\text{-}III(2)} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-4\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{cb}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{cb}(w)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\;U^{ab}(w')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\;U^{ab}(w)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\Big]
\end{aligned}
\tag{9.8}
$$

9.2 From four ρ's: the two-ρ lists of `4rho.tex`

These are the two-ρ terms of `4rho.tex` ("Two ρ contribution", $dN^{(1)}$, $dN^{(2)}$ with c.c.), with the two fixes found earlier (both still needed in this `4rho.tex`):

- **F1:** in the first part, drop $-8N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(x'-w)\,K(y,x',z)$;
- **F2:** in the second part, $dN^{(1)}$, term 2 has one index four times. The correct term is

$$
3f^{abc}f^{deg}\rho^{b'}(x')\rho^{c}(y)U^{ag}(y)U^{db'}(w')U^{eb}(z).
$$

Collected (added):

$$
\left.\frac{d^3N}{d^3k}\right|_{4\rho\text{-lists}} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
$$

$$
\begin{aligned}
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(w)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^m(x')\,\rho^e(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^m(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{ha}(w)\,U^{hl}(x')\,U^{mb}(z)\,U^{md}(x)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^l(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w)\,U^{lh}(x')\,U^{da}(x)\,\rho^l(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{hd}(x)\,\rho^e(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x')\Big]
\end{aligned}
\tag{9.9}
$$

9.3 From reordering the three and four ρ's

The commutator terms of the three-ρ terms of Sec. 8 and of the four-ρ terms. They are your lists of `Evolution_terms.tex`, Sec. 2, which were checked row by row against these rows (0 differing kernel monomials in every row, `Final_evolution_check.pdf`), here with $\log\frac{\vee}{\Lambda}\to\log\frac{k^+}{\Lambda}$ and coloured.

From $4\,\rho's$

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^a(x')\,\rho^c(x)\\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w)\,U^{ca}(x')\,\rho^c(x')\,\rho^b(x)\\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ca}(x)\,\rho^b(x')\,\rho^c(x)\\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)
\end{aligned}
$$

$$
\begin{aligned}
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^a(x)\,\rho^c(x')\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{dh}(w')\,U^{hb}(w)\,\rho^e(x')\,\rho^c(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w)\,U^{dh}(x')\,\rho^e(x')\,\rho^c(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w')\,U^{dh}(x)\,\rho^c(x')\,\rho^e(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{lc}(x)\,U^{ad}(x')\,\rho^h(x')\,\rho^l(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{lc}(x')\,U^{eb}(w)\,U^{ad}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{dh}(w')\,U^{hb}(w)\,\rho^a(x)\,\rho^e(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hb}(w)\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hb}(w')\,U^{dh}(x')\,\rho^a(x)\,\rho^e(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(w')\,U^{dh}(w)\,\rho^a(x)\,\rho^e(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(w')\,U^{dc}(x')\,U^{al}(x)\,\rho^l(x)\,\rho^h(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ec}(x')\,U^{db}(w)\,U^{al}(x)\,\rho^h(x')\,\rho^l(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\Big]
\end{aligned}
\tag{9.10}
$$

From three ρ's

Group I (with c.c.)

Row I Only 4 ρ's

Row II

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Big[\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x')\,U^{ce}(x)\,\rho^l(x')\,\rho^h(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^l(x)\,\rho^h(x')\\
&+2\,N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^e(x')\,\rho^c(x)\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^h(x')\,\rho^l(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ah}(w')\,\rho^c(x)\,\rho^e(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x')\,U^{cl}(x)\,\rho^l(x)\,\rho^h(x')\Big]
\end{aligned}
\tag{9.11}
$$

Row III We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^b(x)\,\rho^c(x') \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^l(x)\,\rho^h(x') \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^l(x')\,\rho^h(x) \\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^l(x)\,\rho^h(x') \\
-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^l(x)\,\rho^h(x')\Bigg]
\end{aligned}
\tag{9.12}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(w)\,\rho^l(x')\,\rho^h(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(w')\,\rho^l(x')\,\rho^h(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(x')\,\rho^l(x')\,\rho^h(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(w)\,\rho^h(x')\,\rho^l(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^h(x')\,\rho^l(x) \\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x) \\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(w')\,\rho^h(x')\,\rho^l(x) \\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

$$
\begin{aligned}
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{cl}(x)\,U^{be}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x)\,U^{be}(x')\,\rho^{l}(x)\,\rho^{h}(x')\Big]
\end{aligned}
\tag{9.13}
$$

Row IV We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ah}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{al}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{le}(w')\,U^{al}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{cb}(x')\,\rho^{c}(x')\,\rho^{a}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bd}(x')\,U^{lc}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ah}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)
\end{aligned}
$$

$$+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$$
$$-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^{b}(x)\,\rho^{c}(x')\Big] \tag{9.14}$$

$$\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\Big[$$
$$-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$$
$$+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$$
$$-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$$
$$+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$$
$$-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{da}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$$
$$+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^{b}(x)\,\rho^{c}(x')$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ea}(w')\,U^{bl}(x)\,U^{dc}(x')\,\rho^{l}(x)\,\rho^{h}(x')$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ah}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')$$
$$+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\Big] \tag{9.15}$$

Row V

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\Big[$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)$$
$$-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')$$
$$+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')$$
$$-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')$$
$$-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)$$
$$+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')$$
$$-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')$$
$$+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')$$
$$+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\Big] \tag{9.16}$$

Group II

Row I

$$\frac{d^3N_{\mathrm{NLO}}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\Big[$$
$$\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)$$
$$+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{al}(w')\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ah}(w)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{bc}(w)\,\rho^{c}(x)\,\rho^{a}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ce}(x')\,U^{al}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{bm}(z)\,U^{md}(x)\,U^{ha}(w)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{bc}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{bc}(w)\,\rho^{c}(x)\,\rho^{a}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hm}(x)\,U^{ld}(x')\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{db}(z)\,U^{al}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(w')\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^{e}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^{e}(x')\,\rho^{m}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ce}(x')\,U^{ad}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ah}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{lb}(z)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{bl}(z)\,U^{lm}(x)\,U^{ha}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ca}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{hd}(w)\,\rho^{e}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{ah}(w')\,U^{bl}(z)\,U^{hd}(x')\,U^{lm}(x)\,\rho^{e}(x')\,\rho^{m}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ce}(x')\,U^{lb}(z)\,U^{ad}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\Bigg]
\end{aligned}
\tag{9.17}
$$

Row II Only 4 ρ's

Row III Only 4 ρ's

Row IV Only 2 ρ's

Group III (with c.c.)

Row I We write it in two parts:

$$
\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^{l}(x')\,\rho^{e}(x)\\
-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^{m}(x')\,\rho^{e}(x)\\
+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hd}(x')\,U^{lm}(w)\,U^{la}(w')\,\rho^{e}(x')\,\rho^{m}(x)\\
&+\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^{h}(x')\,\rho^{c}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{le}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^{c}(x')\,\rho^{h}(x)\\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{md}(x)\,U^{mb}(z)\,U^{ah}(w)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hd}(x)\,U^{lm}(w')\,U^{al}(w)\,\rho^{m}(x')\,\rho^{e}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{le}(x)\,U^{md}(w)\,U^{lb}(z)\,U^{ma}(w')\,\rho^{c}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{be}(x)\,U^{ad}(w)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{db}(z)\,U^{ae}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ld}(x')\,U^{me}(w')\,U^{am}(w)\,\rho^{h}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{mb}(z)\,U^{ah}(w')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(w)\,U^{hb}(z)\,U^{la}(w')\,\rho^{e}(x')\,\rho^{m}(x)\Bigg]
\end{aligned}
\tag{9.18}
$$

$$
\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} &= \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w)\,U^{ma}(w')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{bh}(z)\,U^{hl}(x)\,U^{md}(w')\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{lb}(z)\,U^{ah}(w')\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(w)\,U^{hb}(z)\,U^{ma}(w')\,\rho^{l}(x)\,\rho^{e}(x')
\end{aligned}
$$

$$- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(x)\,U^{ad}(w)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\Big] \tag{9.19}$$

Row II We write it in two parts:

$$\begin{aligned}
\frac{d^3N_{1\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{al}(w)\,\rho^{m}(x')\,\rho^{e}(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x) \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(w)\,\rho^{b}(x')\,\rho^{c}(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{al}(w')\,\rho^{m}(x)\,\rho^{e}(x') \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x') \\
&-\frac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(w')\,\rho^{b}(x)\,\rho^{c}(x') \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\frac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x') \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{ad}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{e}(x')\,\rho^{c}(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x) \\
&-\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{c}(x')\,\rho^{e}(x) \\
&+\frac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{lm}(x')\,U^{bh}(z)\,U^{la}(w)\,\rho^{m}(x')\,\rho^{e}(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ca}(x)\,\rho^{b}(x')\,\rho^{c}(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{eb}(z)\,U^{cd}(x)\,\rho^{l}(x')\,\rho^{h}(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ea}(w')\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ah}(w')\,\rho^{c}(x')\,\rho^{e}(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x)\,U^{db}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{be}(z)\,U^{da}(w)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{eb}(z)\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{lm}(x)\,U^{bh}(z)\,U^{la}(w')\,\rho^{m}(x)\,\rho^{e}(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x') \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ca}(w')\,\rho^{b}(x)\,\rho^{c}(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{db}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x') \\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w)\,\rho^{l}(x)\,\rho^{e}(x') \\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x') \\
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^{b}(x)\,\rho^{e}(x')\Big]
\end{aligned} \tag{9.20}$$

$$\begin{aligned}
\frac{d^3N_{2\,\mathrm{NLO}}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\Big[\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')
\end{aligned}$$

$$
\begin{aligned}
&+\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{am}(w')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\frac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^{e}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w)\,\rho^{l}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{md}(x')\,U^{bh}(z)\,U^{ma}(w')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ah}(w')\,\rho^{e}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^{c}(x')\,\rho^{a}(x)\Big]
\end{aligned}
\tag{9.21}
$$

Row III

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\Big[\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{eb}(z)\,U^{ad}(x)\,U^{cl}(w')\,\rho^{l}(x')\,\rho^{h}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bd}(z)\,U^{ae}(x')\,U^{cl}(w)\,\rho^{l}(x)\,\rho^{h}(x')\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{lb}(z)\,U^{ad}(x')\,U^{ce}(w')\,\rho^{l}(x)\,\rho^{h}(x')\\
-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{bl}(z)\,U^{ae}(x')\,U^{cd}(w)\,\rho^{h}(x')\,\rho^{l}(x)\\
-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\Big]
\end{aligned}
\tag{9.22}
$$

Row IV Only 4 ρ's

Row V

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k}=\frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\log\Big[\frac{k^+}{\Lambda}\Big]\int_{x,x',z,w,w'}e^{-ik(w'-w)}\Big[\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(x)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(w')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{cd}(w)\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{cd}(w)\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{ce}(w')\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{ce}(w')\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{h}(x')\,\rho^{l}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(w)\,U^{bd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(x)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(w')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x')\,U^{ce}(w')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(w)\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cd}(w)\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\Big]
\end{aligned}
\tag{9.23}
$$

Row VI

$$
\begin{aligned}
\frac{d^3N_{\rm NLO}}{d^3k} = \frac{1}{(2\pi)^3}\,\frac{g^4}{16\pi^5}\,\frac{1}{k^+}\,\log\left[\frac{k^+}{\Lambda}\right]\int_{x,x',z,w,w'} e^{-ik(w'-w)}\,\Bigg[\\
\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{be}(z)\,U^{cl}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{eb}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{bd}(z)\,U^{ce}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{cd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{eb}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(x')\,U^{cl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bd}(z)\,U^{cl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{bl}(z)\,U^{ce}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w-z)\,U^{ab}(z)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
-\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(x)\,U^{cd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
+\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{bl}(z)\,U^{cd}(x')\,\rho^{h}(x')\,\rho^{l}(x)\\
+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\Bigg]
\end{aligned}
\tag{9.24}
$$

9.4 The sum

Kernel structure by kernel structure, the colour structures of all terms above are reduced to an independent basis. The sum minus JIMWLK⊗LO is then:

kernel structure	class	independent colour structures	differences
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	charge–charge	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	charge–charge	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	13	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w,z)$	JIMWLK only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w',w',z)$	JIMWLK only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(w,w',z)$	JIMWLK only	9	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	JIMWLK only	14	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	JIMWLK only	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w,z)$	JIMWLK only	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',w',z)$	JIMWLK only	14	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	14	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w,z)$	JIMWLK only	12	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,w',z)$	JIMWLK only	12	0

Without the `4rho.tex` lists, 18 of 152 colour structures would differ (in $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$, $\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$). The sum, collected, is JIMWLK⊗LO; in this form it reads

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{JIMWLK}\otimes\text{LO}} = \mathcal{N}\log\frac{k^+}{\Lambda}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^m(x')\,\rho^e(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^m(x')\,\rho^e(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^l(x')\,\rho^e(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ad}(z)\,U^{bh}(x)\,U^{el}(x')\,\rho^l(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{ah}(x)\,U^{dl}(w')\,U^{eb}(z)\,\rho^l(x')\,\rho^c(x) \\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^l(x)\,\rho^e(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^l(x)\,\rho^e(x') \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x) \\
&+2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x) \\
&-2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{eb}(x')\,U^{lh}(x)\,U^{da}(z)\,\rho^l(x)\,\rho^c(x') \\
&-2\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(z)\,U^{lc}(x')\,U^{be}(x)\,\rho^h(x)\,\rho^l(x') \\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{be}(z)\,U^{ch}(w)\,\rho^l(x')\,\rho^d(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(w')\,U^{lb}(w)\,\rho^m(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(w')\,U^{le}(w)\,\rho^m(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w)\,U^{lm}(x')\,U^{lb}(w)\,\rho^m(x')\,\rho^c(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w-z)^2}\;f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w)\,U^{lm}(x')\,U^{le}(w)\,\rho^m(x')\,\rho^c(x) \\
&+2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^c(x')\,\rho^b(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{be}(z)\,U^{ch}(w')\,\rho^d(x')\,\rho^l(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;\tfrac{1}{(w'-z)^2}\;f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(w)\,\rho^c(x')\,\rho^m(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; f^{abc}\,f^{dea}\,U^{hd}(z)\,U^{he}(w')\,U^{lb}(w')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(w'-z)^2}\; f^{abc}\,f^{deb}\,U^{hd}(z)\,U^{ha}(w')\,U^{le}(w')\,U^{lm}(x)\,\rho^c(x')\,\rho^m(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w')\,U^{cb}(w)\,\rho^a(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{la}(w')\,U^{be}(z)\,U^{cd}(w)\,\rho^l(x')\,\rho^h(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{la}(w)\,U^{be}(z)\,U^{cd}(w')\,\rho^h(x')\,\rho^l(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^c(x')\,\rho^l(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; U^{ab}(w')\,U^{bc}(x)\,\rho^a(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ah}(w')\,\rho^b(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(w'-z)\; f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{be}(z)\,U^{cl}(w')\,\rho^l(x')\,\rho^h(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^l(x')\,\rho^h(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; U^{ab}(w)\,U^{bc}(x')\,\rho^c(x')\,\rho^a(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ah}(w)\,\rho^b(x)\,\rho^e(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x'-z)\cdot\mathcal{K}(w-z)\; f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{bd}(z)\,U^{cl}(w)\,\rho^l(x)\,\rho^h(x')\Big]
\end{aligned}
\tag{9.25}
$$

This is $\mathcal{N}\log\frac{k^+}{\Lambda}\times\mathcal{J}$, Eq. (11.1), before the charge average. Its terms (green / purple): 28 / 20.

10 The two-ρ terms $\propto \log\frac{\vee}{k^+}$: BFKL

The coefficient of $\log\frac{\vee}{k^+}$ of every row is its $\log\frac{\vee}{k^+}$ list of Secs. 4–6.

- It is zero for Group I Rows II (part 1) and V, and for $\mathcal{D}$.
- For the four-ρ terms, their three-ρ remainders and the `4rho.tex` lists it equals their $\log\frac{k^+}{\Lambda}$ terms.
- Group III Rows III, V, VI: the terms (B); Group II Row IV: H_4.

The two-ρ terms (genuine and from reordering the three- and four-ρ terms), row by row (orange / purple: 168 / 162):

Group I, Row II (three ρ;).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|_{I.II}^{\log\frac{\vee}{k^+}} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
&- \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')\\
&+ \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&- \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^c(x)\,\rho^e(x')\Big]
\end{aligned}
\tag{10.1}
$$

Group I, Row III (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|_{I.III}^{\log\frac{\vee}{k^+}} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\; U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\; U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

$$
\begin{aligned}
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\Big]
\end{aligned}
\tag{10.2}
$$

Group I, Row IV (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{I.IV} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)
\end{aligned}
$$

$$
\begin{aligned}
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{le}(w')\,U^{la}(w)\,U^{mb}(z)\,U^{md}(x')\,\rho^{h}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(w)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{10.3}
$$

Group II, Row I (three ρ).

$$
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.I}=\mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'}e^{-ik\cdot(w'-w)}\Big[
$$

$$
\begin{aligned}
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{bde}\, U^{hd}(w')\, U^{ha}(w)\, \rho^{e}(x')\, \rho^{c}(x)\\
&-N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{deh}\, U^{al}(w)\, U^{cd}(x')\, U^{be}(z)\, \rho^{h}(x')\, \rho^{l}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{dce}\, U^{hl}(x)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^{e}(x')\, \rho^{l}(x)\\
&-N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; U^{ab}(w')\, U^{ac}(w)\, \rho^{c}(x)\, \rho^{b}(x')\\
&+N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{ce}(x)\, U^{bl}(z)\, \rho^{l}(x')\, \rho^{h}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{hd}(x)\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(x')\, \rho^{m}(x')\, \rho^{e}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{bde}\, U^{ha}(w')\, U^{hd}(w)\, \rho^{c}(x')\, \rho^{e}(x)\Bigg]
\end{aligned}
\tag{10.4}
$$

Group III, Row I (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.I} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Bigg[\\
&+\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x)\\
&-\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\; U^{ab}(w')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{hl}(x')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^{l}(x')\, \rho^{e}(x)\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^{l}(x')\, \rho^{e}(x)\\
&+\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x)\\
&-\tfrac{1}{2}\, N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\; U^{ab}(w)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x)\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{dce}\, U^{hl}(x)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^{e}(x')\, \rho^{l}(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{dce}\, U^{hl}(w)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^{e}(x')\, \rho^{l}(x)\\
&+N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{ac}(x')\, \rho^{b}(x)\, \rho^{c}(x')\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{cl}(x)\, U^{bd}(z)\, \rho^{l}(x)\, \rho^{h}(x')\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{cl}(x')\, U^{be}(z)\, \rho^{l}(x')\, \rho^{h}(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{cl}(x)\, U^{bd}(x')\, \rho^{l}(x)\, \rho^{h}(x')\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{cl}(x')\, U^{be}(x)\, \rho^{l}(x')\, \rho^{h}(x)\\
&-\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{le}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{md}(x')\, \rho^{h}(x')\, \rho^{c}(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ld}(w)\, U^{la}(w')\, U^{mb}(z)\, U^{me}(x)\, \rho^{c}(x')\, \rho^{h}(x)\\
&-N_c\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x)\\
&+\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ae}(w')\, U^{cd}(x')\, U^{bl}(x)\, \rho^{h}(x')\, \rho^{l}(x)\\
&-\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ad}(w)\, U^{ce}(x')\, U^{bl}(x)\, \rho^{l}(x)\, \rho^{h}(x')\\
&+\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(x)\, \rho^{m}(x)\, \rho^{e}(x')\\
&-\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{hd}(x')\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(x)\, \rho^{m}(x)\, \rho^{e}(x')\\
&-\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(x)\, \rho^{m}(x)\, \rho^{e}(x')\\
&+\tfrac{1}{2}\, \mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{dce}\, U^{hd}(w)\, U^{ha}(w')\, U^{lb}(z)\, U^{lm}(x)\, \rho^{m}(x)\, \rho^{e}(x')\\
&+N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; U^{ab}(x')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x)\\
&-N_c\, \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; U^{ab}(w')\, U^{ac}(x)\, \rho^{b}(x')\, \rho^{c}(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{hl}(x')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^{l}(x')\, \rho^{e}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\, \tfrac{1}{(x-z)^2}\; f^{abc}\, f^{cde}\, U^{hl}(w')\, U^{ha}(w)\, U^{mb}(z)\, U^{md}(x)\, \rho^{l}(x')\, \rho^{e}(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{ea}(w)\, U^{db}(z)\, U^{hl}(x')\, \rho^{l}(x')\, \rho^{c}(x)\\
&+ \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(x')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(x)\, \rho^{m}(x)\, \rho^{e}(x')\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{le}(w')\, U^{la}(w)\, U^{mb}(z)\, U^{md}(x')\, \rho^{h}(x')\, \rho^{c}(x)\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{cde}\, U^{hd}(w')\, U^{ha}(w)\, U^{lb}(z)\, U^{lm}(x)\, \rho^{m}(x)\, \rho^{e}(x')\\
&+N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; U^{ab}(x)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x)\\
&-N_c\, \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; U^{ab}(w)\, U^{ac}(x')\, \rho^{c}(x')\, \rho^{b}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{dce}\, U^{hl}(x)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^{e}(x')\, \rho^{l}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\, \tfrac{1}{(x'-z)^2}\; f^{abc}\, f^{dce}\, U^{hl}(w)\, U^{ha}(w')\, U^{mb}(z)\, U^{md}(x')\, \rho^{e}(x')\, \rho^{l}(x)\\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\, f^{deh}\, U^{da}(w')\, U^{eb}(z)\, U^{hl}(x)\, \rho^{c}(x')\, \rho^{l}(x)
\end{aligned}
$$

$$
\begin{aligned}
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{deh}\,U^{ld}(w)\,U^{la}(w')\,U^{mb}(z)\,U^{me}(x)\,\rho^{c}(x')\,\rho^{h}(x)\\
&+ \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\; f^{abc}\,f^{dce}\,U^{hd}(w)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{10.5}
$$

Group III, Row II (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.II} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+\tfrac{1}{2}\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&-\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\;\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&+N_c\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\;\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x')\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-2\,N_c\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,U^{lb}(z)\,U^{lm}(x)\,\rho^{m}(x)\,\rho^{e}(x')\\
&-2\;\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{dce}\,U^{hl}(x)\,U^{ha}(w')\,U^{mb}(z)\,U^{md}(x')\,\rho^{e}(x')\,\rho^{l}(x)\\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-2\,N_c\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,U^{lb}(z)\,U^{lm}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+2\;\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{c}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{10.6}
$$

Group III, Row III (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.III} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x) \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x') \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x) \\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{10.7}
$$

Group III, Row V (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.V} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(z)\,\rho^h(x')\,\rho^l(x) \\
&- \mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^l(x)\,\rho^h(x') \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(w)\,U^{cd}(x')\,U^{be}(z)\,\rho^h(x')\,\rho^l(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x) \\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&- \mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x)\,U^{bl}(z)\,\rho^l(x')\,\rho^h(x) \\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^l(x')\,\rho^h(x)\Big]
\end{aligned}
\tag{10.8}
$$

Group III, Row VI (three ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.VI} = \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^l(x')\,\rho^h(x) \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^b(x')\,\rho^c(x) \\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x) \\
&-2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(z)\,U^{ac}(x')\,\rho^c(x')\,\rho^b(x)
\end{aligned}
$$

$$
\begin{aligned}
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+2\,N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x)\,U^{cd}(x')\,U^{be}(z)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ca}(z)\,\rho^{b}(x)\,\rho^{c}(x')\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{10.9}
$$

Group II, Row I (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.I^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^{c}(x)\,\rho^{l}(x')\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ea}(w)\,U^{db}(z)\,U^{hl}(x')\,\rho^{l}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{eb}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^{c}(x')\,\rho^{e}(x)\Big]
\end{aligned}
\tag{10.10}
$$

Group II, Row IV (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{II.IV^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;\rho^{a}(x)\,\rho^{a}(x')\\
&-2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&-2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x)\,\rho^{a}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{10.11}
$$

Group III, Row III, second part (genuine two ρ).

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{III.III^{\rm gen}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x')\,\rho^{c}(x)\,\rho^{l}(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{da}(w')\,U^{be}(z)\,U^{hl}(x)\,\rho^{c}(x')\,\rho^{l}(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x)\,\rho^{l}(x)\,\rho^{h}(x')
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{eb}(z)\,U^{lc}(x')\,\rho^{l}(x')\,\rho^{h}(x)\\
&-4\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{bc}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x')\,\rho^{a}(x)\,\rho^{c}(x')\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{cb}(x)\,\rho^{a}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^{c}(x)\,\rho^{d}(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^{c}(x')\,\rho^{d}(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^{d}(x)\,\rho^{c}(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(z)\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^{d}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{10.12}
$$

From four ρ's: remainders d^3N_a, d^3N_b.

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{4\rho,\;a\text{-}b} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\tfrac{1}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\;U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(x')\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(x')\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{e}(x')\,\rho^{c}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{dbe}\,U^{hd}(w)\,U^{ha}(w')\,\rho^{c}(x)\,\rho^{e}(x')\\
&-N_c\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ae}(w')\,U^{cd}(x')\,U^{bl}(x)\,\rho^{h}(x')\,\rho^{l}(x)\\
&-\tfrac{1}{2}\,\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\;f^{abc}\,f^{deh}\,U^{ad}(w)\,U^{ce}(x')\,U^{bl}(x)\,\rho^{l}(x)\,\rho^{h}(x')\Big]
\end{aligned}
\tag{10.13}
$$

From four ρ's: the two-ρ lists of `4rho.tex`.

$$
\begin{aligned}
\frac{d^3N}{d^3k}\Big|^{\log\frac{\vee}{k^+}}_{4\rho\text{-lists}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(x')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;f^{abc}\,f^{deh}\,U^{al}(w')\,U^{ce}(x)\,U^{bd}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\;U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)
\end{aligned}
$$

$$
\begin{aligned}
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hl}(x')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hl}(w')\,U^{ha}(w)\,U^{mb}(z)\,U^{md}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(w)\,U^{bl}(x)\,U^{dl}(z)\,U^{mh}(w')\,\rho^{m}(x')\,\rho^{e}(x)\\
&+3\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(z)\,U^{dl}(w)\,U^{lm}(w')\,U^{hb}(x)\,\rho^{m}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hd}(w)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{ah}(z)\,U^{bl}(x)\,U^{dh}(x)\,U^{ml}(x')\,\rho^{m}(x')\,\rho^{e}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hd}(x)\,U^{hl}(w')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hd}(x)\,U^{hl}(x')\,U^{ma}(z)\,U^{mb}(x)\,\rho^{l}(x')\,\rho^{e}(x)\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,f^{abc}f^{cde}\,U^{hb}(w)\,U^{hl}(x')\,U^{ma}(x)\,U^{md}(z)\,\rho^{l}(x')\,\rho^{e}(x)\\
&-2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+2\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}f^{cde}\,U^{hb}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}f^{cde}\,U^{hb}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{md}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}f^{cde}\,U^{hd}(x')\,U^{hl}(w)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,f^{abc}f^{cde}\,U^{hd}(x')\,U^{hl}(x)\,U^{ma}(z)\,U^{mb}(x')\,\rho^{l}(x)\,\rho^{e}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^{b}(x')\,\rho^{e}(x)\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{ae}(x')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{ad}(x)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+\tfrac{3}{2}\,N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,U^{ab}(x)\,U^{ac}(x')\,\rho^{b}(x)\,\rho^{c}(x')\\
&-\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{ae}(w')\,U^{cl}(x)\,U^{bd}(z)\,\rho^{l}(x)\,\rho^{h}(x')\\
&+\tfrac{1}{2}\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(z)\,\rho^{l}(x')\,\rho^{h}(x)\\
&-\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{deh}\,U^{ad}(w)\,U^{cl}(x')\,U^{be}(x)\,\rho^{l}(x')\,\rho^{h}(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^{e}(x')\,\rho^{c}(x)\Big]
\end{aligned}
\tag{10.14}
$$

kernel structure	class	independent colour structures	differences
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x,z)$	charge–charge	12	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x',x',z)$	charge–charge	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	13	0
$\mathcal{K}(x'-w)\cdot\mathcal{K}(x'-w')\;K(x,x',z)$	charge–charge	14	0
$\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$	BFKL only	14	0
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x,z)$	BFKL only	10	0
$\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\;K(x,x',z)$	BFKL only	13	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x',x',z)$	BFKL only	10	0
$\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\;K(x,x',z)$	BFKL only	13	0

The sum. Collected, the sum is BFKL⊗LO, Eq. (11.2) before the charge average (orange / purple: 23 / 12):

$$
\begin{aligned}
\left.\frac{d^3N}{d^3k}\right|_{\text{BFKL}\otimes\text{LO}} &= \mathcal{N}\log\frac{\vee}{k^+}\int_{x,x',z,w,w'} e^{-ik\cdot(w'-w)}\Big[\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^{b}(x')\,\rho^{c}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(x)\,\rho^{b}(x')\,\rho^{c}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(w')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ac}(x')\,\rho^{c}(x')\,\rho^{b}(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^{b}(x)\,\rho^{e}(x')
\end{aligned}
$$

$$
\begin{aligned}
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dce}\,U^{hd}(x)\,U^{ha}(w')\,\rho^b(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{cde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(x'-w')\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{ade}\,U^{hb}(x)\,U^{hd}(x')\,\rho^e(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x)\,\rho^e(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dbe}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,N_c\,\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,\rho^a(x)\,\rho^a(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dbe}\,U^{ha}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x)\,\rho^d(x')\\
&+\mathcal{K}(w-z)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dbe}\,U^{ha}(w')\,U^{eh}(z)\,\rho^c(x')\,\rho^d(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ac}(w)\,\rho^b(x')\,\rho^c(x)\\
&+N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(w')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-N_c\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\tfrac{1}{(x-z)^2}\,U^{ab}(x')\,U^{ca}(z)\,\rho^b(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{bde}\,U^{hd}(w')\,U^{ha}(w)\,\rho^e(x')\,\rho^c(x)\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{cde}\,U^{hd}(x')\,U^{ha}(w)\,\rho^b(x)\,\rho^e(x')\\
&+2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{ade}\,U^{hb}(w')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x'-w')\cdot\mathcal{K}(w-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{ade}\,U^{hb}(x')\,U^{eh}(z)\,\rho^d(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w')\,U^{ac}(w)\,\rho^c(x)\,\rho^b(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ac}(w')\,\rho^b(x)\,\rho^c(x')\\
&+N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(w)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-N_c\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\tfrac{1}{(x'-z)^2}\,U^{ab}(x)\,U^{ca}(z)\,\rho^b(x)\,\rho^c(x')\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{bde}\,U^{ha}(w')\,U^{hd}(w)\,\rho^c(x')\,\rho^e(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{bde}\,U^{hd}(x)\,U^{ha}(w')\,\rho^c(x')\,\rho^e(x)\\
&-2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dae}\,U^{hb}(w)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\\
&+2\,\mathcal{K}(x-w)\cdot\mathcal{K}(w'-z)\,\mathcal{K}(x-z)\cdot\mathcal{K}(x'-z)\,f^{abc}f^{dae}\,U^{hb}(x)\,U^{eh}(z)\,\rho^d(x')\,\rho^c(x)\Big]
\end{aligned}
\tag{10.15}
$$

11 The targets

In both formulas δY is the evolution interval: $\delta Y = \log\frac{k^+}{\Lambda}$ for JIMWLK and $\delta Y = \log\frac{\vee}{k^+}$ for BFKL. With $\langle\rho^a\rho^b\rangle = \delta^{ab}\langle\rho^f\rho^f\rangle/(N_c^2-1)$ and $\int_{w,w'} e^{-ik\cdot(w'-w)}$, $\mathcal{N}\,\delta Y \leftrightarrow \frac{\alpha_s^2\,\delta Y}{8\pi^6(N_c^2-1)}$.

11.1 JIMWLK⊗LO: your formula, corrected (M2)

The JIMWLK formula of your last message is the one checked before (your App. F): $M = (U^\dagger(x)-U^\dagger(t))(U(y)-U(t))$ in every term of $dN^{(1)}$, $dN^{(3)}$, $M = (U^\dagger(y)-U^\dagger(t))(U(x)-U(t))$ in every term of $dN^{(2)}$, $dN^{(4)}$, and $dN^{(5)}$ with an overall minus sign. (In your message the signs in front of lines 2–4 of $dN^{(5)}$ came through as bullets; they are read as in your App. F, $-,-,+$, as below.) As sent, it differs from JIMWLK⊗LO, i.e. from the two-ρ sum of Sec. 9, by 0.22–0.47 (relative; six configurations, symmetric and general U). Corrected as below, it agrees to 2.1×10^{-15}. With $P \equiv \frac{\alpha_s^2\delta Y}{8\pi^6(N_c^2-1)}$ and $\int \equiv \int_{x,y,t}\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')\langle\rho^f(x)\rho^f(y)\rangle$, the rule is: in a term $\mathcal{K}(u,v,t)\,\mathrm{Tr}\{T^eU^\dagger(v)U(u)T^e\,M\}$,

$$M = (U^\dagger(u)-U^\dagger(t))(U(v)-U(t)),$$

i.e. M sits at the two points the colour charges act on. As sent, M was always $(U^\dagger(x)-U^\dagger(t))(U(y)-U(t))$ in $dN^{(1)}$, $dN^{(3)}$ and $(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))$ in $dN^{(2)}$, $dN^{(4)}$. The corrected formula:

$$
\begin{aligned}
\frac{dN^{(1)}}{d^2w\,d^2w'\,dy} = P\int\Big[&\\
&\mathcal{K}(y,x,t)\mathrm{Tr}\{T^eU^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(w',x,t)\mathrm{Tr}\{T^eU^\dagger(x)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(y,w,t)\mathrm{Tr}\{T^eU^\dagger(w)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\}\\
&+\mathcal{K}(w',w,t)\mathrm{Tr}\{T^eU^\dagger(w)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\}\Big],\\
\frac{dN^{(2)}}{d^2w\,d^2w'\,dy} = P\int\Big[&
\end{aligned}
$$

$$
\begin{aligned}
&\mathcal{K}(x,y,t)\mathrm{Tr}\{T^e U^\dagger(x)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(x,w',t)\mathrm{Tr}\{T^e U^\dagger(x)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(w,y,t)\mathrm{Tr}\{T^e U^\dagger(w)U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(w)-U(t))\}\\
&+\mathcal{K}(w,w',t)\mathrm{Tr}\{T^e U^\dagger(w)U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w)-U(t))\}\Big],
\end{aligned}
$$

$$
\begin{aligned}
\frac{dN^{(3)}}{d^2w d^2w' dy} = &-P\int\Big[\mathcal{K}(y,y,t)\mathrm{Tr}\{T^e(U^\dagger(x)-U^\dagger(w))U(y)T^e(U^\dagger(y)-U^\dagger(t))(U(y)-U(t))\}\\
&-\mathcal{K}(w',w',t)\mathrm{Tr}\{T^e(U^\dagger(x)-U^\dagger(w))U(w')T^e(U^\dagger(w')-U^\dagger(t))(U(w')-U(t))\}\Big],
\end{aligned}
$$

$$
\begin{aligned}
\frac{dN^{(4)}}{d^2w d^2w' dy} = &-P\int\Big[\mathcal{K}(x,x,t)\mathrm{Tr}\{T^e U^\dagger(x)(U(y)-U(w'))T^e(U^\dagger(x)-U^\dagger(t))(U(x)-U(t))\}\\
&-\mathcal{K}(w,w,t)\mathrm{Tr}\{T^e U^\dagger(w)(U(y)-U(w'))T^e(U^\dagger(w)-U^\dagger(t))(U(w)-U(t))\}\Big],
\end{aligned}
$$

$$
\begin{aligned}
\frac{dN^{(5)}}{d^2w d^2w' dy} = &-+P\int\Big[\\
&\mathcal{K}(y,y,t)\mathrm{Tr}\{(U^\dagger(y)-U^\dagger(t))U(y)T^d\}\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))U(y)T^d\}\\
&-\mathcal{K}(w',w',t)\mathrm{Tr}\{(U^\dagger(w')-U^\dagger(t))U(w')T^d\}\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))U(w')T^d\}\\
&-\mathcal{K}(x,x,t)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(t))U(x)T^d\}\mathrm{Tr}\{U^\dagger(x)(U(y)-U(w'))T^d\}\\
&+\mathcal{K}(w,w,t)\mathrm{Tr}\{(U^\dagger(w)-U^\dagger(t))U(w)T^d\}\mathrm{Tr}\{U^\dagger(w)(U(y)-U(w'))T^d\}\Big],
\end{aligned}
\tag{11.1}
$$

and $\mathcal{J}=\sum_{i=1}^{5} dN^{(i)}$ (per unit δY). After the correction, $dN^{(2)}$ is identical to $dN^{(1)}$ term by term. Numerically, at random points with random adjoint Wilson lines, $\sum_i dN^{(i)}$ equals JIMWLK⊗LO to 2×10^{-15}; as sent it differs by 0.22–0.47 (relative).

11.2 BFKL⊗LO: your formula (correct as sent)

$$
\begin{aligned}
\left.\frac{dN}{d^2w\,d^2w'\,dy}\right|_{\mathrm{BFKL}} = &-\frac{\alpha_s^2 N_c\,\delta Y}{8\pi^6(N_c^2-1)}\int_{x,y,z}\langle\rho^f(x)\rho^f(y)\rangle\Big[\\
&\mathcal{K}(x-w)\cdot\mathcal{K}(y-w')K_D(x,y,z)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(y)-U(w'))\}\\
&-2\,\mathcal{K}(z-w)\cdot\mathcal{K}(z-w')K(x,y,z)\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(z)-U(w'))\}\\
&+\mathcal{K}(z-w)\cdot\mathcal{K}(y-w')\big(2K(x,y,z)-K(x,x,z)\big)\mathrm{Tr}\{(U^\dagger(z)-U^\dagger(w))(U(y)-U(w'))\}\\
&+\mathcal{K}(x-w)\cdot\mathcal{K}(z-w')\big(2K(x,y,z)-K(y,y,z)\big)\mathrm{Tr}\{(U^\dagger(x)-U^\dagger(w))(U(z)-U(w'))\}\Big],
\end{aligned}
\tag{11.2}
$$

with $K_D(x,y,z)=K(x,x,z)-2K(x,y,z)+K(y,y,z)$. It agrees with BFKL⊗LO to 8×10^{-15}.

Your BFKL formula (Eq. (LOBFKL) of your message) is correct as written: with $K_D=K(x,x,z)-2K(x,y,z)+K(y,y,z)$, $\delta Y=\log\frac{\nu}{k^+}$ and the singlet projection above, it equals BFKL⊗LO, i.e. the two-ρ sum of Sec. 10, to 8.2×10^{-15} (six configurations).

12 Poles

With $E(r)=\frac{2}{\epsilon}+2\gamma_E+\log\frac{r^2\mu^2}{4}$:

- The poles $\times\log\frac{k^+}{\Lambda}$ of Group I Row II, Group I Row V 2b UV and Group II Row IV are in the ratio $1:-2:1$ and cancel (8.2×10^{-16}, three configurations; `ll_checks.py poles`).
- The $\log\frac{\nu}{k^+}$ list has two terms with a pole, (i) and (ii) of Group II Row IV; they cancel exactly. So the BFKL terms are pole free.

13 Checks

Scripts in `full_check/` unless stated:

check	result
every step of your Group files (`step_check.py`)	Sec. 2
every equation of `4rho.tex` vs. the checked file (`four_file_check.py`)	Sec. 2
every evaluable term = its log $\frac{k^+}{\Lambda}$, log $\frac{\vee}{k^+}$ and remaining parts, exactly in Λ, $\vee$ (`ll_checks.py split`; 79 terms, 3 configurations, 2 sets of $k^+, \vee, \Lambda$)	1.5×10^{-14}
remaining parts: exact remainder $\to$ its $\Lambda \to 0$ form	3.5×10^{-8} at $\Lambda = 10^{-9}$
$\frac{1}{\Lambda}$ lines of Group I Row III add up to zero; your two red lines	3.0×10^{-16}; 0
the three parts equal those of the earlier files, row by row (`same_lists.py`)	0
earlier `BFKL.tex` + `Other_terms.tex` = new rows − JIMWLK; their BFKL list = this one; remaining parts (`rows_vs_files.py`)	2.5×10^{-9}; 8.8×10^{-17}; 3.4×10^{-16}
exact identities of line 1 (`bfkl_rows/exact_check.py`)	$\leq 3.7 \times 10^{-15}$; (B): $\leq 3.4 \times 10^{-15}$
poles $\times$ log $\frac{k^+}{\Lambda}$ cancel (`ll_checks.py poles`)	8.2×10^{-16}
three ρ's $\propto$ log $\frac{k^+}{\Lambda}$ / $\propto$ log $\frac{\vee}{k^+}$ cancel (`ll_checks.py three`)	7.7×10^{-16} / 1.5×10^{-16}
the same with the edge kernel instead of $\mathcal{D}$ / without (B)	0.59 / 3.1×10^{-16}
four ρ's cancel (`four_file_check.py`)	1.3×10^{-16}
two-ρ lists of `4rho.tex` (log restored, F1, F2) vs. the checked ones	5.6×10^{-15}
two ρ's $\propto$ log $\frac{k^+}{\Lambda}$ vs JIMWLK⊗LO; $\propto$ log $\frac{\vee}{k^+}$ vs BFKL⊗LO, by colour structure (`final_check/render.py`)	0 of 154; 0 of 109 differ
your JIMWLK formula: as sent / corrected; your BFKL formula (`final_check/targets_check.py`)	0.22–0.47 / 2.1×10^{-15}; 8.2×10^{-15}